Increments, and the problem of the tangent

Two questions from opposite ends of science — the speed of a falling stone and the slope of a curve — that turn out to be the same question.

LESSON 1–2 | 2 × 40 MIN | ALGEBRA 11, §1.1 | P1 · 7.1

LESSON CLOCK

7' | 14' | 16' | 18' | 18' | 7'

Two problems, one answer	7 min
Δx and Δy	14 min
The difference quotient	16 min
Secant to tangent	18 min
Practice	18 min
Homework	7 min

LEARNING OBJECTIVES

- Define the increment of the argument and the increment of the function.
- Compute Δy for a given function and Δx.
- Form the difference quotient and interpret it as an average rate of change.
- Explain why a tangent cannot be found by the two-point method alone.

TEXTBOOK

National	pp. 5–14
Cambridge	Pure Mathematics 1, pp. 122–126
Workbook	P1 Exercise 7A

01 Explanation

Two old problems

The physicist's problem. A stone falls $s = 5t^2$ metres in t seconds. Its average speed over the first 2 seconds is $20 \div 2 = 10$ m/s. But how fast is it moving **at the instant** $t = 2$? Dividing distance by time needs an interval, and an instant has none.

The geometer's problem. The gradient of a straight line is $\dfrac{\text{rise}}{\text{run}}$ between any two of its points. A curve has a different steepness everywhere. What is the gradient **at one point**?

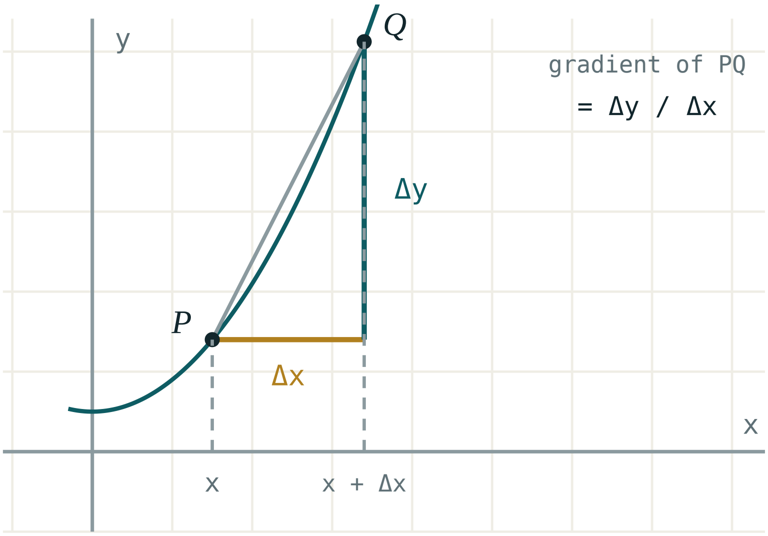
THE SAME QUESTION

Both ask for a ratio at a single point, where the ratio has nothing to divide. Both are answered by the same idea: compute the ratio over a small interval, then shrink the interval.

Increments

Move the input from x_0 to a nearby x. The change in the input is the **increment of the argument**; the change it causes in the output is the **increment of the function**:

$$\Delta x = x - x_0 \quad \Delta y = f(x_0 + \Delta x) - f(x_0)$$



Δx along the axis, Δy up the side. The secant is the hypotenuse of that triangle.

Δx IS ONE SYMBOL

It is not Δ multiplied by x. It cannot be cancelled, split, or treated as a product. It may be positive or negative — the point may move left.

For $y = x^2$ at $x_0 = 3$ with $\Delta x = 0.1$: $\Delta y = 3.1^2 - 3^2 = 9.61 - 9 = 0.61$.

The difference quotient

Their ratio is the **average rate of change** over the interval — and geometrically it is the gradient of the **secant** through the two points:

$$\frac{\Delta y}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

For $y = x^2$ at $x_0 = 3$, expand once and the pattern appears:

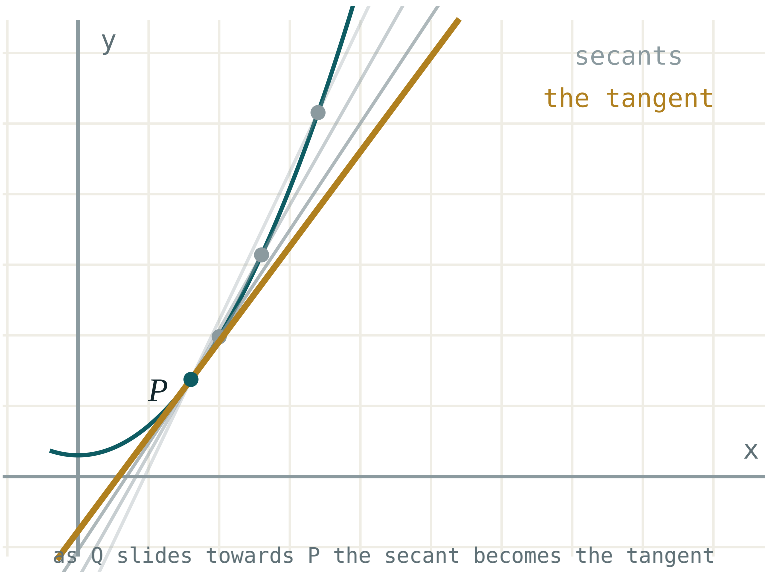
$$\frac{(3 + \Delta x)^2 - 9}{\Delta x} = \frac{6\Delta x + (\Delta x)^2}{\Delta x} = 6 + \Delta x$$

ΔX	ΔY	$\Delta Y/\Delta X$
1	7	7
0.1	0.61	6.1
0.01	0.0601	6.01
0.001	0.006001	6.001

The quotient is walking towards 6 and never arrives. That number is the answer to both problems.

Secant to tangent

Hold A fixed and slide B towards it along the curve. The secant AB turns, and settles into a limiting position: the **tangent** at A.



Three secants and the line they are approaching. The tangent is the limit, not one of them.

YOU CANNOT JUST SET $\Delta X = 0$

That gives $\frac{0}{0}$, which is meaningless. The whole of the next lesson exists to make “approaches without reaching” into something precise.

02 Worked examples

EXAMPLE 1 **For $y = x^2$, find Δy at $x_0 = 4$ with $\Delta x = 0.2$.**

1. $f(4.2) = 17.64$
2. $f(4) = 16$
3. $\Delta y = 1.64$
4. $$\frac{\Delta y}{\Delta x} = \frac{1.64}{0.2} = 8.2$$
   Close to 8.

ANSWER

$\Delta y = 1.64,\ \dfrac{\Delta y}{\Delta x} = 8.2$

EXAMPLE 2 **Simplify $\dfrac{\Delta y}{\Delta x}$ for $y = 3x^2 - x$ at a general x.**

1. $\Delta y = 3(x + \Delta x)^2 - (x + \Delta x) - (3x^2 - x)$
2. $= 6x\Delta x + 3(\Delta x)^2 - \Delta x$
   The $3x^2$ and x terms cancel.
3. $$\frac{\Delta y}{\Delta x} = 6x + 3\Delta x - 1$$

ANSWER

$6x + 3\Delta x - 1$

EXAMPLE 3 **A stone falls $s = 5t^2$. Find its average speed on $[2, 2.01]$.**

1. $s(2.01) = 5 \times 4.0401 = 20.2005$
2. $s(2) = 20$
3. $\Delta s = 0.2005$, $\Delta t = 0.01$
4. 20.05 m/s.

   Approaching 20.

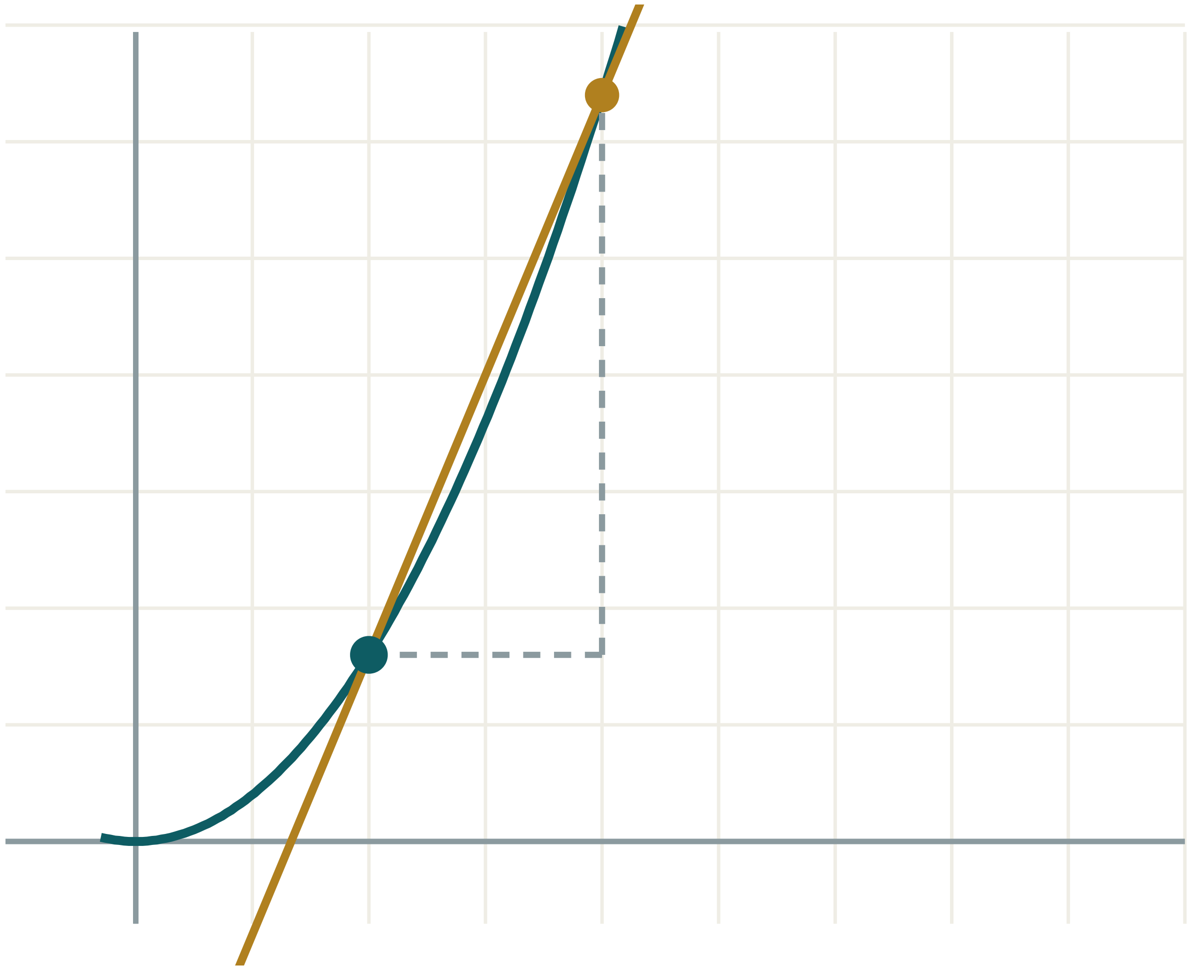
ANSWER

20.05 m/s

03 Interactive model

Slide h towards zero and read the gradient of the secant as it settles.

Secant to tangent



x **2** | h **2** | Δy / Δx **2.4** | f ′(x) exactly **1.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Increment of the argument	Argument orttirmasi	Приращение аргумента
Increment of the function	Funksiya orttirmasi	Приращение функции
Difference quotient	Ayirmalar nisbati	Разностное отношение
Average rate of change	O‘rtacha o‘zgarish tezligi	Средняя скорость изменения
Instantaneous rate	Oniy tezlik	Мгновенная скорость
Secant	Kesuvchi	Секущая
Tangent	Urinma	Касательная
Gradient	Burchak koeffitsienti	Угловой коэффициент
Limiting position	Limit holat	Предельное положение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Δx means:

- Δ times x
- the change in x
- the derivative
- a small positive number

$\frac{\Delta y}{\Delta x}$ is the gradient of:

- the tangent
- the secant
- the normal
- the axis

For $y = x^2$ at $x = 3$, $\frac{\Delta y}{\Delta x}$ simplifies to:

- 6
- $6 + \Delta x$
- $3 + \Delta x$
- 9

Setting $\Delta x = 0$ directly gives:

- the answer
- $\frac{0}{0}$
- zero
- infinity

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $y = x^2$, $x_0 = 2$, $\Delta x = 1$. Find Δy

   ANSWER 5

2. $y = x^2$, $x_0 = 2$, $\Delta x = 0.1$. Find Δy

   ANSWER 0.41

3. $y = 3x$, any x_0, $\Delta x = 2$. Find Δy

   ANSWER 6

4. $y = 5$. Find Δy

   ANSWER 0

5. $y = x$. Find $\dfrac{\Delta y}{\Delta x}$

   ANSWER 1

6. $s = 5t^2$. Average speed on $[1, 2]$

   ANSWER 15 m/s

7. What does the secant become as $\Delta x \to 0$?

   ANSWER the tangent

Medium · the standard question

7 PROBLEMS

1. $y = x^2$. Simplify $\frac{\Delta y}{\Delta x}$

ANSWER $2x + \Delta x$

2. $y = x^3$. Simplify $\frac{\Delta y}{\Delta x}$

ANSWER $3x^2 + 3x\Delta x + (\Delta x)^2$

3. $y = 2x^2 + 1$. Simplify $\frac{\Delta y}{\Delta x}$

ANSWER $4x + 2\Delta x$

4. $y = \frac{1}{x}$. Simplify $\frac{\Delta y}{\Delta x}$

ANSWER $-\frac{1}{x(x + \Delta x)}$

5. $s = 5t^2$. Average speed on $[2, 2.1]$

ANSWER 20.5 m/s

6. $y = x^2 - 4x$. $\frac{\Delta y}{\Delta x}$ at $x = 1$

ANSWER $-2 + \Delta x$

7. What number is $2x + \Delta x$ approaching?

ANSWER $2x$

Hard · stretch and reason

7 PROBLEMS

1. $y = \sqrt{x}$. Simplify $\frac{\Delta y}{\Delta x}$

ANSWER $\frac{1}{\sqrt{x + \Delta x} + \sqrt{x}}$

2. $y = \frac{1}{x^2}$. Simplify $\frac{\Delta y}{\Delta x}$

ANSWER $-\frac{2x + \Delta x}{x^2(x + \Delta x)^2}$

3. Explain why $\frac{\Delta y}{\Delta x}$ cannot be evaluated at $\Delta x = 0$

ANSWER Both parts vanish; $\frac{0}{0}$ is undefined

4. $y = x^2$. For which Δx is the secant gradient at $x = 3$ equal to 6.5?

ANSWER $\Delta x = 0.5$

5. A ball: $h = 20t - 5t^2$. Average velocity on $[1, 1 + \Delta x]$

ANSWER $10 - 5\Delta x$

6. When is that average velocity zero?

ANSWER $\Delta x = 2$

7. Interpret the sign of $\frac{\Delta y}{\Delta x}$ for a decreasing function

ANSWER It is negative

07 Homework

Homework — 5 tasks

Task 5 needs a table, like the one in the lesson.

1. $y = x^2$. Find Δy at $x_0 = 5$ for $\Delta x = 1, 0.1, 0.01$.

2. $y = x^3 - x$. Simplify $\frac{\Delta y}{\Delta x}$ at a general x.
3. $s = 4t^2 + t$. Find the average speed on $[3, 3.1]$.
4. Explain in three sentences why the tangent cannot be found from two points on the curve.
5. Tabulate $\frac{\Delta y}{\Delta x}$ for $y = x^2$ at $x = 2$ with $\Delta x = 1, 0.5, 0.1, 0.01$ and say what it approaches.

The limit of a function

Making "approaches without reaching" precise, and the four techniques that evaluate almost every limit you will meet.

LESSON 3–4 | 2 × 40 MIN | ALGEBRA 11, §1.2 | P1 · 7.1

LESSON CLOCK

9' | 12' | 20' | 16' | 18' | 5'

What a limit says	9 min
Substitution, when it works	12 min
The 0/0 form	20 min
Algebra of limits	16 min
Practice	18 min
Homework	5 min

LEARNING OBJECTIVES

- State informally what lim f(x) = L means.
- Evaluate limits by substitution where the function is continuous.
- Resolve a 0/0 limit by factorising, by rationalising, or by cancelling.
- Use the algebra of limits for sums, products and quotients.

TEXTBOOK

National	pp. 15–26
Cambridge	Pure Mathematics 1, pp. 126–129
Workbook	P1 Exercise 7A

01 Explanation

What the notation says

$$\lim_{x \to a} f(x) = L$$

IN WORDS

As x gets close to a — from either side, but never equal to a — the value $f(x)$ gets and stays as close to L as you like.

The value $f(a)$ is **irrelevant**. The function need not even be defined there. $\frac{x^2 - 1}{x - 1}$ has no value at $x = 1$, yet its limit there is 2: everywhere except at 1 it equals $x + 1$.

Substitution, and when it fails

For a polynomial, or any function continuous at a, the limit is just the value:

$$\lim_{x \to 2} (x^2 + 3x) = 4 + 6 = 10$$

Substitution fails when it produces an **indeterminate form** — most often $\frac{0}{0}$. That is not an answer; it is a signal that algebra is needed first.

> $\frac{0}{0}$ IS NOT ZERO AND NOT ONE
>
> It means the numerator and denominator both vanish, and their **rate** of vanishing decides the answer. Different pairs give different limits.

Three techniques for $\frac{0}{0}$

SHAPE	TECHNIQUE	EXAMPLE
polynomial over polynomial	factorise and cancel	$\frac{x^2-9}{x-3} \to x+3 \to 6$
a square root appears	multiply by the conjugate	$\frac{\sqrt{x}-2}{x-4} \to \frac{1}{\sqrt{x}+2} \to \frac{1}{4}$
a compound fraction	combine, then cancel	$\frac{\frac{1}{x}-\frac{1}{2}}{x-2} \to -\frac{1}{4}$

All three do the same thing: remove the factor that is making both parts vanish.

$$\frac{x^2-9}{x-3} = \frac{(x-3)(x+3)}{x-3} = x + 3 \ (x \neq 3)$$

The cancellation is legal precisely because $x \neq 3$ throughout the limiting process.

The algebra of limits

If both limits exist, limits pass through the four operations:

$$\lim(f \pm g) = \lim f \pm \lim g \cdot \lim(f \cdot g) = \lim f \cdot \lim g \cdot \lim \frac{f}{g} = \frac{\lim f}{\lim g} \text{ (if } \lim g \neq 0)$$

and $\lim(c \cdot f) = c \cdot \lim f$. These rules are what let you break a complicated limit into pieces — and the condition on the quotient rule is exactly where the $\frac{0}{0}$ cases hide.

WHY THIS LESSON EXISTS

The derivative is defined as one particular limit, of exactly the $\frac{0}{0}$ kind. Every technique here is used again in the next lesson.

02 Worked examples

EXAMPLE 1 **Evaluate $\lim_{x \to 3} \frac{x^2 - x - 6}{x - 3}$.**

1. Substitution gives $\frac{0}{0}$.

   Factorise.

2. $x^2 - x - 6 = (x - 3)(x + 2)$

3. $= x + 2$ for $x \neq 3$.

4. $\to 5$

ANSWER

5

EXAMPLE 2 **Evaluate $\lim_{x \to 0} \dfrac{\sqrt{x+4} - 2}{x}$.**

1. $\dfrac{0}{0}$ — multiply by the conjugate.
2. $\dfrac{(x+4) - 4}{x(\sqrt{x+4} + 2)}$
3. $= \dfrac{1}{\sqrt{x+4} + 2}$
4. $\to \dfrac{1}{4}$

ANSWER

$\dfrac{1}{4}$

EXAMPLE 3 **Evaluate $\lim_{x \to 2} \dfrac{x^3 - 8}{x^2 - 4}$.**

1. $x^3 - 8 = (x-2)(x^2 + 2x + 4)$
2. $x^2 - 4 = (x-2)(x+2)$
3. $= \dfrac{x^2 + 2x + 4}{x + 2}$
4. $\to \dfrac{12}{4} = 3$

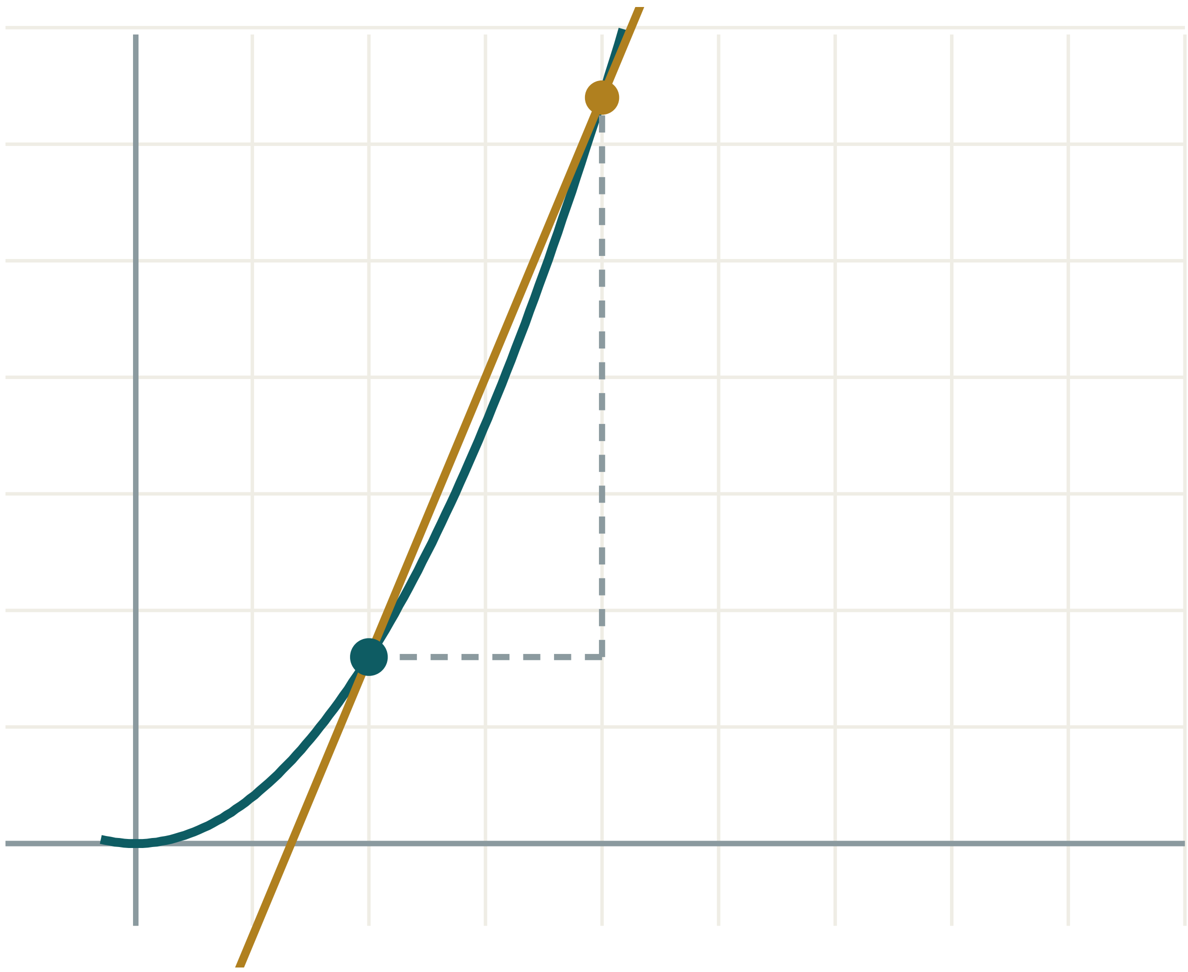
ANSWER

3

03 Interactive model

Tabulate the function either side of the point before doing any algebra.

Watching a limit



x **2** | h **2** | Δy / Δx **2.4** | f ′(x) exactly **1.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Limit	Limit	Предел
Tends to	Intiladi	Стремится к
Continuous function	Uzluksiz funksiya	Непрерывная функция
Point of discontinuity	Uzilish nuqtasi	Точка разрыва
Indeterminate form	Aniqmaslik	Неопределённость
One-sided limit	Bir tomonlama limit	Односторонний предел
Removable discontinuity	Bartaraf etiluvchi uzilish	Устранимый разрыв
Conjugate expression	Qo‘shma ifoda	Сопряжённое выражение
Infinity	Cheksizlik	Бесконечность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\lim_{x \to 2} (x^2 + 1)$ is:

- 4
- 5
- 3
- undefined

$\frac{0}{0}$ means:

- the limit is 0
- the limit is 1
- more work is needed
- the limit does not exist

$\lim_{x \to 1} \frac{x^2 - 1}{x - 1}$ is:

- 0
- 1
- 2
- undefined

To handle $\frac{\sqrt{x} - 3}{x - 9}$ you should:

- substitute
- factorise
- multiply by the conjugate
- give up

The value $f(a)$ affects $\lim_{x \to a} f(x)$:

- always
- never
- sometimes
- only if defined

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\lim_{x \to 1} (3x + 2)$

ANSWER 5

2. $\lim_{x \to 0} (x^2 - 4)$

ANSWER -4

3. $\lim_{x \to 2} \dfrac{x^2 - 4}{x - 2}$

ANSWER 4

4. $\lim_{x \to 3} \dfrac{x^2 - 9}{x - 3}$

ANSWER 6

5. $\lim_{x \to 5} 7$

ANSWER 7

6. $\lim_{x \to 1} \dfrac{x - 1}{x^2 - 1}$

ANSWER 0.5

7. Is $\dfrac{0}{0}$ an answer?

ANSWER no

Medium · the standard question

7 PROBLEMS

1. $\lim_{x \to -2} \dfrac{x^2 + 3x + 2}{x + 2}$

ANSWER -1

2. $\lim_{x \to 4} \dfrac{\sqrt{x} - 2}{x - 4}$

ANSWER $\dfrac{1}{4}$

3. $\lim_{x \to 0} \dfrac{\sqrt{x + 9} - 3}{x}$

ANSWER $\dfrac{1}{6}$

4. $\lim_{x \to 2} \dfrac{x^3 - 8}{x - 2}$

ANSWER 12

5. $\lim_{x \to 1} \dfrac{x^3 - 1}{x^2 - 1}$

ANSWER 1.5

6. $\lim_{x \to 3} \dfrac{\frac{1}{x} - \frac{1}{3}}{x - 3}$

ANSWER $-\dfrac{1}{9}$

7. $\lim_{x \to 0} \dfrac{(2 + x)^2 - 4}{x}$

ANSWER 4

Hard · stretch and reason

7 PROBLEMS

1. $\lim_{x \to 1} \dfrac{x^4 - 1}{x - 1}$

ANSWER 4

2. $\lim_{x \to 0} \dfrac{\sqrt{1 + x} - \sqrt{1 - x}}{x}$

ANSWER 1

3. $\lim_{x \to 2} \dfrac{x^3 - 3x^2 + 4}{x^2 - 4}$

ANSWER $\dfrac{3}{4}$

4. $\lim_{x \to a} \dfrac{x^2 - a^2}{x - a}$

ANSWER $2a$

5. $\lim_{x \to a} \dfrac{x^n - a^n}{x - a}$

ANSWER na^{n-1}

6. Find k so that $\dfrac{x^2 + kx - 6}{x - 2}$ has a finite limit at 2

ANSWER $k = 1$

7. Give two functions with $\dfrac{0}{0}$ at 0 and different limits

ANSWER $\dfrac{x}{x} \to 1$ and $\dfrac{x^2}{x} \to 0$

07 Homework

Homework — 5 tasks

Show the algebra. A limit written down with no working scores nothing.

1. Evaluate $\lim_{x \to 4} \dfrac{x^2 - 16}{x - 4}$.
2. Evaluate $\lim_{x \to 0} \dfrac{\sqrt{x + 16} - 4}{x}$.
3. Evaluate $\lim_{x \to 3} \dfrac{x^3 - 27}{x^2 - 9}$.
4. Explain in three sentences why $\dfrac{0}{0}$ is called indeterminate.
5. Evaluate $\lim_{x \to a} \dfrac{x^3 - a^3}{x - a}$ and check it against na^{n-1}.

The derivative of a function

The definition, the notation, and what the number actually measures in geometry and in physics.

LESSON 5–6 | 2 × 40 MIN | ALGEBRA 11, §1.3 | P1 · 7.1–7.2

LESSON CLOCK

9' | 19' | 14' | 11' | 20' | 7'

The definition	9 min
First principles, three times	19 min
Two interpretations	14 min
Notation, and where it fails	11 min
Practice	20 min
Homework	7 min

LEARNING OBJECTIVES

- State the definition of the derivative as a limit of the difference quotient.
- Differentiate from first principles for a polynomial and a simple root.
- Interpret f′(x) as a gradient and as a rate of change.
- Use the three standard notations for the derivative.

TEXTBOOK

National	pp. 27–38
Cambridge	Pure Mathematics 1, pp. 122–133
Workbook	P1 Exercise 7A, 7B

01 Explanation

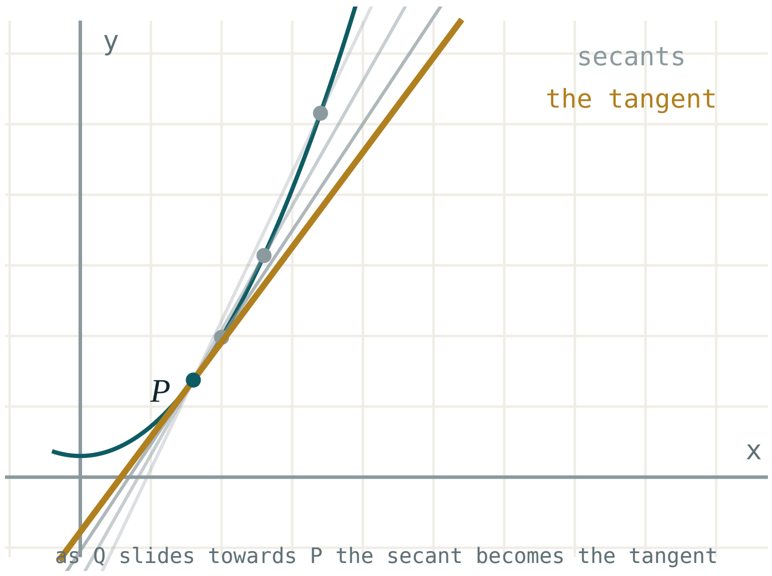
The definition

THE DERIVATIVE OF F AT X

$$f'(x) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h}$$

when that limit exists. The function f is then **differentiable** at x.

Everything from the last two lessons is now in one line: the increments give the quotient, and the limit turns the secant into the tangent.



As h shrinks the secant becomes the tangent, and its gradient becomes f′(x).

Three derivatives from first principles

1. $f(x) = x^2$:

$$\frac{(x+h)^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h \to 2x$$

2. $f(x) = x^3$:

$$\frac{(x+h)^3 - x^3}{h} = 3x^2 + 3xh + h^2 \to 3x^2$$

3. $f(x) = \frac{1}{x}$:

$$\frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \frac{-1}{x(x+h)} \to -\frac{1}{x^2}$$

THE PATTERN

$x^2 \to 2x$, $x^3 \to 3x^2$, $x^{-1} \to -x^{-2}$. In every case the index comes down in front and drops by one. That is the rule proved in the next lesson.

What the number means

CONTEXT	$F(X)$	$F'(X)$
geometry	height of the curve	gradient of the tangent
motion	displacement $s(t)$	velocity $v(t)$
motion, again	velocity $v(t)$	acceleration $a(t)$
economics	total cost $C(q)$	marginal cost
biology	population $P(t)$	growth rate

For the falling stone $s = 5t^2$: $s' = 10t$, so at $t = 2$ the speed is 20 m/s — the number the table in Lesson 1–2 was walking towards.

Notation, and where the derivative fails to exist

Three notations, all standard, all meaning the same thing:

$$f'(x) \cdot y' \cdot \frac{\mathrm{dy}}{\mathrm{dx}}$$

The Leibniz form $\frac{\mathrm{dy}}{\mathrm{dx}}$ keeps the increments visible and is preferred in physics and in the chain rule.

NOT EVERY FUNCTION HAS A DERIVATIVE EVERYWHERE

$f(x) = |x|$ has no derivative at 0: approaching from the left the quotient is -1, from the right $+1$. The two one-sided limits disagree, so no tangent exists — the graph has a **corner**. A vertical tangent, as in $y = \sqrt[3]{x}$ at 0, also fails.

02 Worked examples

EXAMPLE 1 **Differentiate $f(x) = 3x^2 - 5x$ from first principles.**

1. $f(x + h) = 3(x + h)^2 - 5(x + h)$
2. $f(x + h) - f(x) = 6xh + 3h^2 - 5h$
3. $\frac{\cdots}{h} = 6x + 3h - 5$
4. $h \to 0$

ANSWER

$f'(x) = 6x - 5$

EXAMPLE 2 **Differentiate $f(x) = \sqrt{x}$ from first principles.**

1. $\frac{\sqrt{x + h} - \sqrt{x}}{h}$

   Multiply by the conjugate.
2. $= \frac{h}{h(\sqrt{x + h} + \sqrt{x})}$
3. $= \frac{1}{\sqrt{x + h} + \sqrt{x}}$
4. $\to \frac{1}{2\sqrt{x}}$

ANSWER

$f'(x) = \frac{1}{2\sqrt{x}}$

EXAMPLE 3 **A body moves with $s = t^3 - 6t^2$ metres. Find its velocity and acceleration at $t = 4$.**

1. $v = s' = 3t^2 - 12t$
2. $v(4) = 48 - 48 = 0$

   Momentarily at rest.
3. $a = v' = 6t - 12$
4. $a(4) = 12$

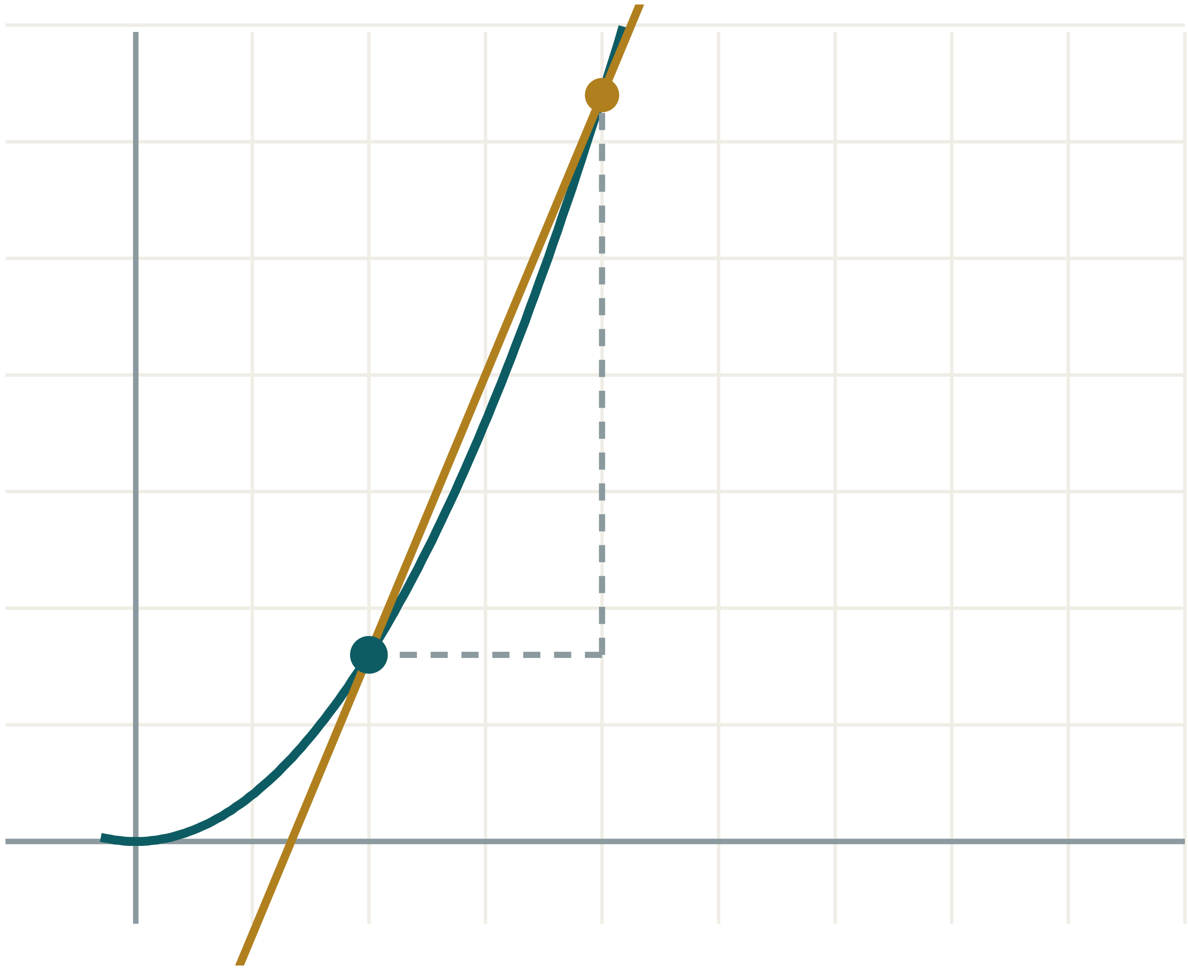
ANSWER

$v = 0$ m/s, $a = 12$ m/s²

03 Interactive model

Set h to 1, then 0.1, then 0.01 and read the gradient converging.

The derivative as a limit



x **2** | h **2** | Δy / Δx **2.4** | f ′(x) exactly **1.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Derivative	Hosila	Производная
Differentiation	Differensiallash	Дифференцирование
From first principles	Ta’rif bo‘yicha	По определению
Gradient function	Burchak koeffitsienti funksiyasi	Функция углового коэффициента
Differentiable	Differensiallanuvchi	Дифференцируемая
Instantaneous velocity	Oniy tezlik	Мгновенная скорость
Leibniz notation	Leybnits belgilanishi	Обозначение Лейбница
Corner point	Sinish nuqtasi	Угловая точка
Rate of change	O‘zgarish tezligi	Скорость изменения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The derivative is the limit of:

- the function
- the secant gradient
- the tangent
- the increment

From first principles, $(x^3)'$ is:

- $3x$
- $3x^2$
- x^2
- $3x^3$

$f'(x)$ for $f(x) = \frac{1}{x}$ is:

- $\frac{1}{x^2}$
- $-\frac{1}{x^2}$
- $\ln x$
- $-x$

$|x|$ at $x = 0$:

- has derivative 0
- has derivative 1
- has no derivative
- has derivative −1

If $s(t)$ is displacement, $s'(t)$ is:

- distance
- velocity
- acceleration
- time

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. From first principles, $(x^2)'$

   ANSWER $2x$

2. From first principles, $(5x)'$

   ANSWER 5

3. From first principles, $(7)'$

   ANSWER 0

4. $f(x) = x^2$; find $f'(3)$

   ANSWER 6

5. $f(x) = x^3$; find $f'(2)$

   ANSWER 12

6. $s = 5t^2$; find $v(2)$

   ANSWER 20 m/s

7. Name three notations for the derivative

   ANSWER $f'(x), y', \frac{\mathrm{dy}}{\mathrm{dx}}$

Medium · the standard question

7 PROBLEMS

1. First principles: $(2x^2 + x)'$

   ANSWER $4x + 1$

2. First principles: $(x^2 - 3x + 1)'$

   ANSWER $2x - 3$

3. First principles: $\left(\frac{1}{x}\right)'$

   ANSWER $-\frac{1}{x^2}$

4. First principles: $(\sqrt{x})'$

   ANSWER $\frac{1}{2\sqrt{x}}$

5. $s = t^3 - 6t^2$; find $v(4)$

   ANSWER 0

6. Same; find $a(4)$

   ANSWER 12

7. Where does $|x - 2|$ fail to be differentiable?

   ANSWER $x = 2$

Hard · stretch and reason

7 PROBLEMS

1. First principles: $(x^4)'$

ANSWER $4x^3$

2. First principles: $\left(\frac{1}{x^2}\right)'$

ANSWER $-\frac{2}{x^3}$

3. First principles: $\left(\frac{1}{\sqrt{x}}\right)'$

ANSWER $-\frac{1}{2x\sqrt{x}}$

4. Show $f(x) = |x|$ has one-sided derivatives ± 1 at 0

ANSWER Quotient is $\frac{|h|}{h}$

5. $s = t^3 - 3t^2 + 2t$; when is the body at rest?

ANSWER $t = 1 \pm \frac{1}{\sqrt{3}}$

6. A differentiable function is always continuous. Give a continuous function that is not differentiable

ANSWER $|x|$ at 0

7. From first principles find $(x^{-3})'$

ANSWER $-3x^{-4}$

07 Homework

Homework — 5 tasks

Tasks 1–3 must be done from the definition, not from a rule.

1. From first principles, differentiate $f(x) = 4x^2 - 7x + 2$.

2. From first principles, differentiate $f(x) = \frac{2}{x}$.
3. From first principles, differentiate $f(x) = \sqrt{2x}$.
4. $s = 2t^3 - 9t^2 + 12t$. Find the velocity and the times at which the body is at rest.
5. Explain, with a sketch, why $y = |x + 1|$ has no derivative at $x = -1$.

The rules of differentiation

Power, constant multiple, sum, product and quotient — five rules that replace first principles for good.

LESSON 7–9 | 3 × 40 MIN | ALGEBRA 11, §1.4 | P1 · 7.2–7.4

LESSON CLOCK

12'	14'	23'	23'	29'	19'

The power rule	12 min
Sums and multiples	14 min
The product rule	23 min
The quotient rule	23 min
Practice	29 min
Homework and consolidation	19 min

LEARNING OBJECTIVES

- Apply the power rule to positive, negative and fractional indices.
- Differentiate sums and constant multiples.
- Apply the product rule and the quotient rule.
- Rewrite an expression into a differentiable form before starting.

TEXTBOOK

National	pp. 39–54
Cambridge	Pure Mathematics 1, pp. 130–145
Workbook	P1 Exercise 7B–7D

01 Explanation

The power rule

$(x^n)' = n \cdot x^{n-1}$ for every real n

$F(X)$	REWRITE AS	$F'(X)$
x^5	x^5	$5x^4$
$\sqrt{x}$	$x^{1/2}$	$\frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$
$\frac{1}{x^3}$	x^{-3}	$-3x^{-4} = -\frac{3}{x^4}$
$\frac{1}{\sqrt{x}}$	$x^{-1/2}$	$-\frac{1}{2x\sqrt{x}}$
x	x^1	1
7	$7x^0$	0

REWRITE BEFORE YOU DIFFERENTIATE

Every root becomes a fractional index; every fraction with x alone below becomes a negative index. Do that first, and the power rule handles everything.

Multiples and sums

$$(c \cdot f)' = c \cdot f' \quad (f \pm g)' = f' \pm g'$$

Together these differentiate any polynomial term by term:

$$(4x^3 - 7x^2 + 2x - 9)' = 12x^2 - 14x + 2$$

THERE IS NO PRODUCT RULE FOR A CONSTANT

$(5x^3)'$ is $15x^2$, not $(5)'(x^3) + 5(x^3)'$. Constants pass straight through — using the product rule here is not wrong, just wasteful.

The product rule

$$(u \cdot v)' = u'v + uv'$$

THE DERIVATIVE OF A PRODUCT IS NOT THE PRODUCT OF THE DERIVATIVES

Test it: $(x \cdot x)' = (x^2)' = 2x$, but $x' \cdot x' = 1$. The rule is needed.

Method. Write u and v in a column, differentiate each beside it, then assemble. For $y = x^3(2x + 1)$:

	U	V
function	x^3	$2x + 1$
derivative	$3x^2$	2

$$y' = 3x^2(2x + 1) + x^3 \cdot 2 = 6x^3 + 3x^2 + 2x^3 = 8x^3 + 3x^2$$

Check by expanding first: $y = 2x^4 + x^3$, so $y' = 8x^3 + 3x^2$. ✓

The quotient rule

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

THE ORDER IN THE NUMERATOR MATTERS

It is $u'v$ **minus** uv', not the other way round. Swapping them flips the sign of every answer. The product rule is symmetric; the quotient rule is not.

For $y = \dfrac{x^2}{x + 1}$:

$$y' = \frac{2x(x + 1) - x^2 \cdot 1}{(x + 1)^2} = \frac{x^2 + 2x}{(x + 1)^2}$$

OFTEN YOU DO NOT NEED IT

$y = \dfrac{x^3 + 2x}{x}$ is just $x^2 + 2$. Divide through first whenever the denominator is a single term — it is faster and safer.

02 Worked examples

EXAMPLE 1 **Differentiate $y = 3x^4 - \frac{2}{x^2} + 5\sqrt{x}$.**

1. Rewrite: $3x^4 - 2x^{-2} + 5x^{(1/2)}$.
2. $12x^3$
3. $+ 4x^{-3}$

   Note the double negative.
4. $+ \frac{5}{2}x^{(-1/2)}$

ANSWER

$y' = 12x^3 + \frac{4}{x^3} + \frac{5}{2\sqrt{x}}$

EXAMPLE 2 **Differentiate $y = (x^2 + 1)(3x - 4)$ by the product rule.**

1. $u = x^2 + 1,\ u' = 2x$
2. $v = 3x - 4,\ v' = 3$
3. $y' = 2x(3x - 4) + (x^2 + 1)(3)$
4. $= 6x^2 - 8x + 3x^2 + 3 = 9x^2 - 8x + 3$

ANSWER

$9x^2 - 8x + 3$

EXAMPLE 3 **Differentiate $y = \dfrac{2x - 1}{x^2 + 3}$.**

1. $u = 2x - 1,\ u' = 2$

2. $v = x^2 + 3,\ v' = 2x$

3. $$y' = \frac{2(x^2 + 3) - (2x - 1)(2x)}{(x^2 + 3)^2}$$

4. $$= \frac{-2x^2 + 2x + 6}{(x^2 + 3)^2}$$

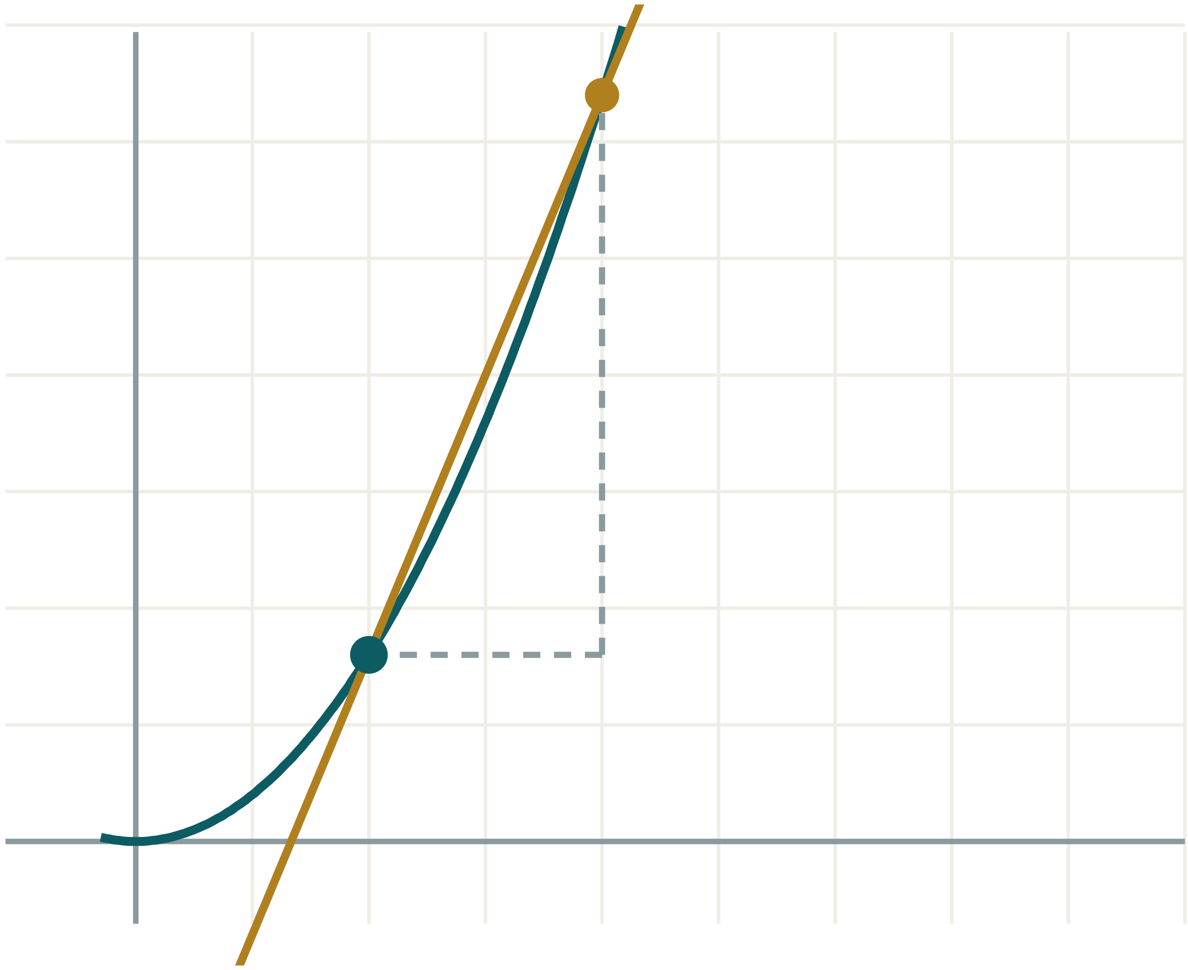
ANSWER

$$\frac{-2x^2 + 2x + 6}{(x^2 + 3)^2}$$

03 Interactive model

Do one product both ways — expand first, then the rule — and confirm they agree.

Checking a rule



x **2** | h **2** | Δy / Δx **2.4** | f ′(x) exactly **1.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Power rule	Daraja qoidasi	Правило степени
Constant multiple rule	O‘zgarmas ko‘paytuvchi qoidasi	Правило постоянного множителя
Sum rule	Yig‘indi qoidasi	Правило суммы
Product rule	Ko‘paytma qoidasi	Правило произведения
Quotient rule	Bo‘linma qoidasi	Правило частного
Fractional index	Kasr ko‘rsatkich	Дробный показатель
Negative index	Manfiy ko‘rsatkich	Отрицательный показатель
Simplify first	Avval soddalashtirish	Сначала упростить
Second derivative	Ikkinchi tartibli hosila	Вторая производная

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$(x^7)'$ is:

- $7x^6$
- $7x^8$
- x^6
- $6x^7$

$(\sqrt{x})'$ is:

- $\frac{1}{2\sqrt{x}}$
- $2\sqrt{x}$
- $\frac{1}{\sqrt{x}}$
- $\frac{1}{2}$

The product rule is:

- $u'v'$
- $u'v + uv'$
- $u'v - uv'$
- $uv' - u'v$

The quotient rule numerator is:

- $u'v + uv'$
- $u'v - uv'$
- $uv' - u'v$
- $u'v'$

$\left(\frac{1}{x^4}\right)'$ is:

- $-\frac{4}{x^5}$
- $\frac{4}{x^5}$
- $-\frac{1}{x^5}$
- $-4x^3$

$(6)'$ is:

- 6
- 1
- 0
- $6x$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(x^6)'$

ANSWER $6x^5$

2. $(3x^2)'$

ANSWER $6x$

3. $(x^3 + x)'$

ANSWER $3x^2 + 1$

4. $(5)'$

ANSWER 0

5. $(4x)'$

ANSWER 4

6. $\left(\frac{1}{x}\right)'$

ANSWER $-\frac{1}{x^2}$

7. $(\sqrt{x})'$

ANSWER $\frac{1}{2\sqrt{x}}$

Medium · the standard question

7 PROBLEMS

1. $(2x^4 - 3x^2 + 7)'$

ANSWER $8x^3 - 6x$

2. $\left(\frac{3}{x^2}\right)'$

ANSWER $-\frac{6}{x^3}$

3. $(x\sqrt{x})'$

ANSWER $\frac{3}{2}\sqrt{x}$

4. $((x + 1)(x - 3))'$

ANSWER $2x - 2$

5. $(x^2(x + 5))'$

ANSWER $3x^2 + 10x$

6. $\left(\frac{x}{x + 1}\right)'$

ANSWER $\frac{1}{(x + 1)^2}$

7. $\left(\frac{x^2 + 1}{x}\right)'$

ANSWER $1 - \frac{1}{x^2}$

Hard · stretch and reason

7 PROBLEMS

1. $((2x - 1)(x^2 + 3))'$

ANSWER $6x^2 - 2x + 6$

2. $\left(\dfrac{2x + 3}{3x - 1}\right)'$

ANSWER $-\dfrac{11}{(3x - 1)^2}$

3. $(x^3\sqrt{x})'$

ANSWER $\dfrac{7}{2}$x^(5/2)

4. $\left(\dfrac{\sqrt{x}}{x + 1}\right)'$

ANSWER $\dfrac{1 - x}{2\sqrt{x}(x + 1)^2}$

5. $(x^2(x - 1)(x + 1))'$

ANSWER $4x^3 - 2x$

6. Find y'' for $y = x^4 - 3x^2$

ANSWER $12x^2 - 6$

7. Find x where $y = x^3 - 3x$ has gradient 9

ANSWER $x = \pm 2$

07 Homework

Homework — 6 tasks

Rewrite roots and fractions as indices before differentiating.

1. Differentiate $y = 5x^4 - \dfrac{3}{x} + 2\sqrt{x}$.
2. Differentiate $y = (3x + 2)(x^2 - 1)$ twice — by expanding and by the product rule.

3. Differentiate $y = \dfrac{x - 2}{x + 2}$.

4. Differentiate $y = x^2\sqrt{x}$.

5. Find y'' for $y = 2x^5 - x^3 + 4x$.

6. Find the points on $y = x^3 - 12x$ where the tangent is horizontal.

The derivative of a composite function

The chain rule — the one rule that makes everything else in calculus usable.

LESSON 10–12 | 3 × 40 MIN | ALGEBRA 11, §1.5 | P1 · 7.5

LESSON CLOCK

12' | 20' | 23' | 23' | 27' | 15'

Why a new rule is needed	12 min
The chain rule	20 min
Practising it	23 min
Mixing with product and quotient	23 min
Related rates	27 min
Homework	15 min

LEARNING OBJECTIVES

- Identify the inner and outer function of a composite expression.
- Apply the chain rule in both notations.
- Combine the chain rule with the product and quotient rules.
- Use the chain rule for related rates of change.

TEXTBOOK

National	pp. 55–70
Cambridge	Pure Mathematics 1, pp. 146–155
Workbook	P1 Exercise 7E

01 Explanation

Why the power rule is not enough

$y = (3x + 1)^5$ is a fifth power, so is it $5(3x + 1)^4$? Test the idea on the easiest case, $y = (3x)^2$. Expanding gives $9x^2$, whose derivative is $18x$ — but $2(3x) = 6x$. The answer is out by a factor of 3: exactly the derivative of the inside.

THE CHAIN RULE

$$(f(g(x)))' = f'(g(x)) \cdot g'(x) \text{ or } \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Differentiate the outside, leaving the inside untouched — then multiply by the derivative of the inside.

Using it

Method. Name the inside u, write y in terms of u, differentiate both, multiply.

y	u	$\frac{dy}{du}$	$\frac{du}{dx}$	$\frac{dy}{dx}$
$(3x+1)^5$	$3x+1$	$5u^4$	3	$15(3x+1)^4$
$\sqrt{x^2+1}$	x^2+1	$\frac{1}{2\sqrt{u}}$	$2x$	$\frac{x}{\sqrt{x^2+1}}$
$\frac{1}{(2x-5)^3}$	$2x-5$	$-3u^{-4}$	2	$-\frac{6}{(2x-5)^4}$
$(x^2-4x)^7$	x^2-4x	$7u^6$	$2x-4$	$7(2x-4)(x^2-4x)^6$

NEVER FORGET THE SECOND FACTOR

Omitting $\frac{du}{dx}$ is the single commonest error in the whole of A-level calculus. Every bracket, every root, every reciprocal needs it.

Mixing the rules

Real expressions need two rules at once. Decide the **outermost** structure first:

EXPRESSION	OUTERMOST STRUCTURE	RULES
$x^2(2x+1)^5$	a product	product, then chain inside
$\frac{(x+1)^3}{x^2}$	a quotient	quotient, then chain
$(x^2(x+1))^4$	a power	chain, then product inside
$\sqrt{x(x+3)}$	a root	chain, then product inside

$$(x^2(2x+1)^5)' = 2x(2x+1)^5 + x^2 \cdot 5(2x+1)^4 \cdot 2 = 2x(2x+1)^4(6x+1)$$

ALWAYS FACTORISE THE ANSWER

The common bracket comes out, and the factorised form is what every later question — stationary points, sign of the gradient — actually needs.

Related rates

The Leibniz form makes the chain rule read as a genuine cancellation, which is why it handles rates that depend on other rates:

$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

Example. A balloon's radius grows at 2 cm/s. How fast is the volume growing when $r = 5$?

$$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dr} = 4\pi r^2 = 100\pi$$

$$\frac{dV}{dt} = 100\pi \times 2 = 200\pi \approx 628\ cm^3/s$$

SUBSTITUTE THE NUMBER LAST

Differentiate symbolically first, then put $r = 5$ in. Substituting first turns the variable into a constant and the derivative into zero.

02 Worked examples

EXAMPLE 1 **Differentiate $y = (4x - 3)^6$.**

1. $u = 4x - 3$
2. $\frac{dy}{du} = 6u^5$
3. $\frac{du}{dx} = 4$
4. $6(4x - 3)^5 \times 4$

ANSWER

$24(4x - 3)^5$

EXAMPLE 2 **Differentiate $y = x\sqrt{x^2 + 5}$.**

1. A product; the second factor needs the chain rule.

2. $$\left(\sqrt{x^2+5}\right)' = \frac{x}{\sqrt{x^2+5}}$$

3. $$y' = \sqrt{x^2+5} + \frac{x^2}{\sqrt{x^2+5}}$$

4. $$= \frac{2x^2+5}{\sqrt{x^2+5}}$$
Over a common denominator.

ANSWER

$$\frac{2x^2+5}{\sqrt{x^2+5}}$$

EXAMPLE 3 **The side of a cube grows at *0.5* cm/s. How fast is the volume growing when the side is *8* cm?**

1. $V = a^3$

2. $$\frac{dV}{da} = 3a^2 = 192$$

3. $$\frac{dV}{dt} = 192 \times 0.5$$

4. $= 96$

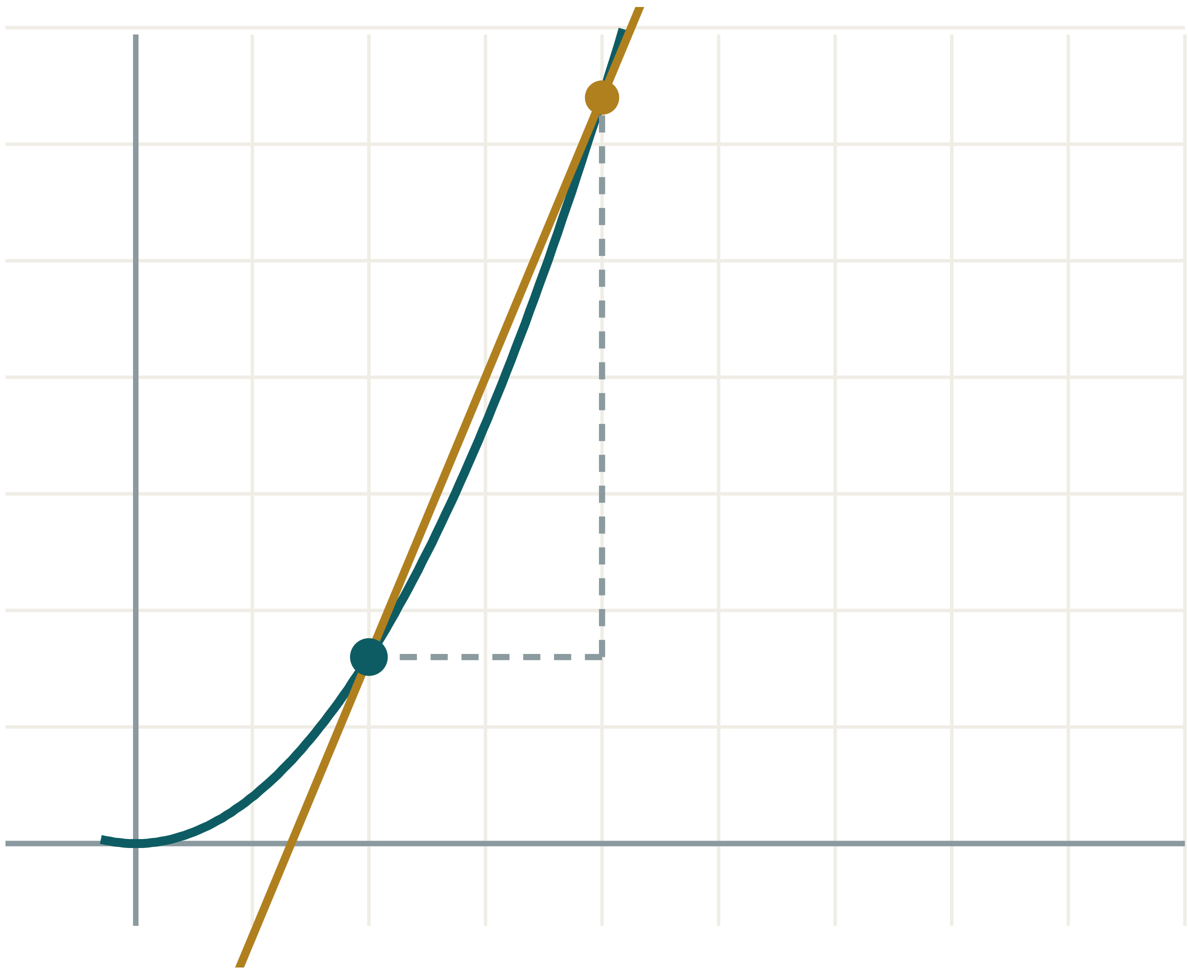
ANSWER

96 cm³/s

03 Interactive model

Differentiate one bracket by expanding and by the chain rule, and check the answers agree.

Composite gradients



x **2** | h **2** | Δy / Δx **2.4** | f ′(x) exactly **1.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Chain rule	Zanjir qoidasi	Правило цепочки
Composite function	Murakkab funksiya	Сложная функция
Inner function	Ichki funksiya	Внутренняя функция
Outer function	Tashqi funksiya	Внешняя функция
Substitution u	Almashtirish	Замена переменной
Related rates	Bog‘liq tezliklar	Связанные скорости
Bracket power	Qavs darajasi	Степень скобки
Intermediate variable	Oraliq o‘zgaruvchi	Промежуточная переменная

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$((2x + 1)^3)'$ is:

- $3(2x + 1)^2$
- $6(2x + 1)^2$
- $2(2x + 1)^2$
- $3(2x + 1)^3$

$(\sqrt{x^2 + 1})'$ is:

- $\dfrac{1}{2\sqrt{x^2 + 1}}$
- $\dfrac{x}{\sqrt{x^2 + 1}}$
- $\dfrac{2x}{\sqrt{x^2 + 1}}$
- $\sqrt{2x}$

The chain rule in Leibniz form is:

- $\dfrac{dy}{dx} = \dfrac{dy}{du} + \dfrac{du}{dx}$
- $\dfrac{dy}{dx} = \dfrac{dy}{du} \cdot \dfrac{du}{dx}$
- $\dfrac{dy}{dx} = \dfrac{du}{dy} \cdot \dfrac{dx}{du}$
- $\dfrac{dy}{du}$

$((x^2 + 3)^4)'$ is:

- $4(x^2 + 3)^3$
- $8x(x^2 + 3)^3$
- $2x(x^2 + 3)^3$
- $4x(x^2 + 3)^3$

For related rates you should substitute the number:

- first
- last
- never
- twice

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $((x + 1)^2)'$

ANSWER $2(x + 1)$

2. $((3x)^4)'$

ANSWER $324x^3$

3. $((2x + 5)^3)'$

ANSWER $6(2x + 5)^2$

4. $((x - 7)^5)'$

ANSWER $5(x - 7)^4$

5. $(\sqrt{2x})'$

ANSWER $\dfrac{1}{\sqrt{2x}}$

6. $((4 - x)^3)'$

ANSWER $-3(4 - x)^2$

7. $((x^2 + 1)^2)'$

ANSWER $4x(x^2 + 1)$

Medium · the standard question

7 PROBLEMS

1. $((3x-2)^7)'$

ANSWER $21(3x-2)^6$

2. $(\sqrt{x^2+9})'$

ANSWER $\dfrac{x}{\sqrt{x^2+9}}$

3. $\left(\dfrac{1}{(x+2)^3}\right)'$

ANSWER $-\dfrac{3}{(x+2)^4}$

4. $((x^2-3x)^4)'$

ANSWER $4(2x-3)(x^2-3x)^3$

5. $(x(x+1)^3)'$

ANSWER $(x+1)^2(4x+1)$

6. $\left(\dfrac{(2x+1)^2}{x}\right)'$

ANSWER $\dfrac{(2x+1)(2x-1)}{x^2}$

7. Radius grows at 3 cm/s; find $\dfrac{dA}{dt}$ for a circle at $r = 4$

ANSWER 24π cm²/s

Hard · stretch and reason

7 PROBLEMS

1. $(x^2\sqrt{x+1})'$

   ANSWER $\dfrac{x(5x+4)}{2\sqrt{x+1}}$

2. $\left(\dfrac{\sqrt{x+1}}{x}\right)'$

   ANSWER $-\dfrac{x+2}{2x^2\sqrt{x+1}}$

3. $(((x+1)^2+3)^4)'$

   ANSWER $8(x+1)((x+1)^2+3)^3$

4. $(\sqrt{x\sqrt{x}})'$

   ANSWER $\dfrac{3}{4}x\text{^}(-1/4)$

5. Balloon: $\dfrac{dr}{dt} = 2$ cm/s. Find $\dfrac{dV}{dt}$ at $r = 10$

   ANSWER 800π cm³/s

6. A ladder 5 m slides: foot moves at 0.4 m/s. Find the top's speed when the foot is 3 m out

   ANSWER 0.3 m/s downwards

7. Show $\left(\dfrac{1}{g(x)}\right)' = -\dfrac{g'(x)}{(g(x))^2}$

   ANSWER Chain rule on g^{-1}

07 Homework

Homework — 6 tasks

Factorise every answer. Task 6 needs the rate set out with its units.

1. Differentiate $y = (5x-1)^4$.

2. Differentiate $y = \sqrt{3x^2 + 2}$.
3. Differentiate $y = \dfrac{1}{(x^2 + 1)^2}$.
4. Differentiate $y = x^3(2x - 1)^4$ and factorise the answer.
5. Differentiate $y = \dfrac{(x + 2)^3}{x}$.
6. A spherical balloon is inflated so that its radius grows at 1.5 cm/s. Find the rate of increase of its volume when $r = 6$ cm.

Control work 1, and work on the mistakes

Limits, first principles and the five rules, in one paper — then an hour repairing what it finds.

LESSON 13–14 | 2 × 40 MIN | ALGEBRA 11, NAZORAT ISHI 1 | P1 · CHAPTER 7 REVIEW

LESSON CLOCK

3 | 35' | 9' | 22' | 11'

Instructions	3 min
The paper	35 min
Self-mark	9 min
Rewrite and classify	22 min
Rule-selection drill	11 min

LEARNING OBJECTIVES

- Differentiate accurately under time.
- Choose the correct rule without being told which.
- Classify each lost mark as careless, method or knowledge.
- Rewrite every wrong solution correctly.

TEXTBOOK

National	pp. 71–74
Cambridge	Pure Mathematics 1, pp. 156–158
Workbook	Control paper A

01 Explanation

The paper — 25 marks, 40 minutes

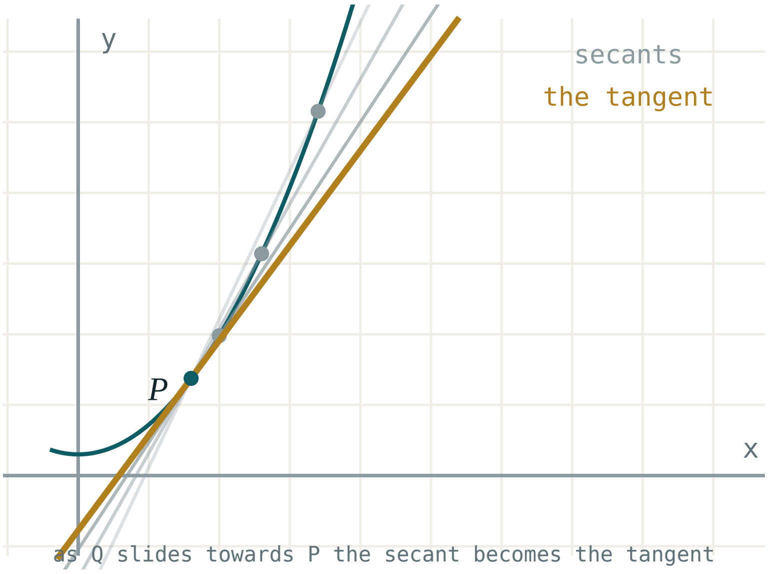
Q	TASK	MARKS	FROM
1	Evaluate $lim_{x \to 3} \frac{x^2 - 9}{x - 3}$ and $lim_{x \to 0} \frac{\sqrt{x + 4} - 2}{x}$	*4*	L3–4
2	Differentiate $f(x) = 2x^2 - 5x$ from first principles	*4*	L5–6
3	Differentiate $y = 3x^4 - \frac{2}{x} + \sqrt{x}$	*4*	L7–9
4	Differentiate $y = (2x + 1)(x^2 - 3)$ by the product rule	*3*	L7–9
5	Differentiate $y = \frac{x}{x^2 + 1}$	*3*	L7–9
6	Differentiate $y = (3x^2 - 1)^5$ and $y = \sqrt{x^2 + 4}$	*4*	L10–12
7	A cube's side grows at 0.2 cm/s. Find $\frac{dV}{dt}$ when the side is 5 cm	*3*	L10–12

Q2 IS MARKED ON THE DEFINITION

A correct answer obtained from the power rule scores **one** mark out of four. “From first principles” means the limit must appear.

The three columns

KIND	IN THIS TOPIC IT LOOKS LIKE	THE REPAIR
careless	a lost sign in the quotient rule, a dropped 2 in $\frac{du}{dx}$	check every answer by re-reading the rule
method	using the product rule where the chain rule was needed	the rule-selection drill below
knowledge	not knowing $(\sqrt{x})'$	learn the six standard derivatives



Question 2 in one picture — the secant, the limit, the tangent.

The rule-selection drill

Twelve expressions on the board. For each, the class calls out only the **rule**, not the answer, in five seconds:

EXPRESSION	RULE
$x^5 - 2x$	power and sum
$(x + 1)^7$	chain
$x^2(x + 1)$	product (or expand)
$\frac{x}{x+1}$	quotient
$\frac{x^2+x}{x}$	divide first, then power
$\sqrt{4x+1}$	chain
$x\sqrt{x+1}$	product then chain
$\frac{1}{(x-2)^4}$	chain

ROWS 5 AND 8 ARE THE TRAPS

Row 5 needs no quotient rule at all; row 8 needs no quotient rule either, once it is written as $(x-2)^{-4}$. Choosing the heavier rule costs time and invites sign errors.

02 Worked examples

EXAMPLE 1 **Model answer, Q2: differentiate $f(x) = 2x^2 - 5x$ from first principles.**

1. $f(x+h) = 2(x+h)^2 - 5(x+h)$
2. $f(x+h) - f(x) = 4xh + 2h^2 - 5h$
3. $\frac{\cdots}{h} = 4x + 2h - 5$
4. $\lim_{h \to 0} = 4x - 5$

ANSWER

$f'(x) = 4x - 5$

EXAMPLE 2 **Model answer, Q5: differentiate $y = \frac{x}{x^2+1}$.**

1. $u = x,\ u' = 1;\ v = x^2 + 1,\ v' = 2x$
2. $\frac{1(x^2+1) - x(2x)}{(x^2+1)^2}$
3. $= \frac{1 - x^2}{(x^2+1)^2}$

ANSWER

$$\frac{1 - x^2}{(x^2+1)^2}$$

EXAMPLE 3 **A learner wrote $((3x^2 - 1)^5)' = 5(3x^2 - 1)^4$. Name the mistake.**

1. The inside derivative is missing.
   A method error.

2. $(3x^2 - 1)' = 6x$

3. Correct: $30x(3x^2 - 1)^4$.

ANSWER

Method error — the chain rule’s second factor was omitted

03 Interactive model

Run the rule-selection drill before the rewrite, not after.

Which rule?

$y = (x^2 + 1)^6$

- power only
- chain
- product
- quotient

$y = x^3(x + 2)$

- chain
- product
- quotient
- none

$y = \frac{x^3 + x}{x}$

- quotient
- divide first
- chain
- product

$y = \frac{1}{(2x + 1)^3}$

- quotient
- chain on $(2x + 1)^{-3}$
- product
- power only

$y = \sqrt{x^2 + x}$

- chain
- product
- quotient
- power only

$((3x^2 - 1)^5)'$ is:

- $5(3x^2 - 1)^4$
- $30x(3x^2 - 1)^4$
- $15x(3x^2 - 1)^4$
- $6x(3x^2 - 1)^4$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Careless error	E’tiborsizlik xatosi	Ошибка по невнимательности
Method error	Usul xatosi	Ошибка в методе
Knowledge gap	Bilim bo‘shlig‘i	Пробел в знаниях
Mark scheme	Baholash sxemasi	Схема оценивания
Working	Yechim yozuvi	Ход решения
Correction	Tuzatish	Исправление
Rule selection	Qoidani tanlash	Выбор правила

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

“From first principles” means:

- use the power rule
- use the limit definition
- use a calculator
- draw a graph

Forgetting $\frac{du}{dx}$ is:

- careless
- a method error
- a knowledge gap
- not an error

$\frac{x^2 + x}{x}$ is best differentiated by:

- the quotient rule
- dividing first
- the product rule
- first principles

The quotient rule numerator is:

- $u'v + uv'$
- $u'v - uv'$
- $uv' - u'v$
- $u'v'$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\lim_{x\to 3} \dfrac{x^2 - 9}{x - 3}$

   ANSWER 6

2. First principles: $(2x^2)'$

   ANSWER $4x$

3. $(3x^4)'$

   ANSWER $12x^3$

4. $\left(\dfrac{2}{x}\right)'$

   ANSWER $-\dfrac{2}{x^2}$

5. $((2x + 1)(x^2 - 3))'$

   ANSWER $6x^2 + 2x - 6$

6. $\left(\dfrac{x}{x^2 + 1}\right)'$

   ANSWER $\dfrac{1 - x^2}{(x^2 + 1)^2}$

7. $((3x^2 - 1)^5)'$

   ANSWER $30x(3x^2 - 1)^4$

Medium · the standard question

7 PROBLEMS

1. $\lim_{x \to 0} \dfrac{\sqrt{x+4} - 2}{x}$

   ANSWER $\dfrac{1}{4}$

2. First principles: $(2x^2 - 5x)'$

   ANSWER $4x - 5$

3. $(3x^4 - \dfrac{2}{x} + \sqrt{x})'$

   ANSWER $12x^3 + \dfrac{2}{x^2} + \dfrac{1}{2\sqrt{x}}$

4. $(\sqrt{x^2 + 4})'$

   ANSWER $\dfrac{x}{\sqrt{x^2 + 4}}$

5. Cube side grows at 0.2 cm/s; $\dfrac{dV}{dt}$ at side 5

   ANSWER 15 cm³/s

6. $(x^2(x + 1)^3)'$

   ANSWER $x(x + 1)^2(5x + 2)$

7. $(\dfrac{2x - 1}{x + 3})'$

   ANSWER $\dfrac{7}{(x + 3)^2}$

Hard · stretch and reason

7 PROBLEMS

1. $\lim_{x \to 2} \dfrac{x^3 - 8}{x^2 - 4}$

   ANSWER 3

2. First principles: $\left(\dfrac{1}{x}\right)'$

   ANSWER $-\dfrac{1}{x^2}$

3. $(x\sqrt{x^2 + 1})'$

   ANSWER $\dfrac{2x^2 + 1}{\sqrt{x^2 + 1}}$

4. $\left(\dfrac{(x + 1)^2}{x - 1}\right)'$

   ANSWER $\dfrac{(x + 1)(x - 3)}{(x - 1)^2}$

5. Find x where $y = (x^2 - 4)^3$ has a horizontal tangent

   ANSWER $x = 0, \pm 2$

6. Sphere: $\dfrac{dr}{dt} = 0.5$ cm/s; find $\dfrac{dV}{dt}$ at $r = 4$

   ANSWER 32π cm³/s

7. Find y'' for $y = (2x + 1)^4$

   ANSWER $48(2x + 1)^2$

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. It is not optional.

1. Rewrite in full every control-work question that lost a mark, with the “I lost this because …” sentence.

2. Five problems from the section your knowledge column was heaviest in.
3. Differentiate $y = x^2\sqrt{2x + 1}$ and factorise the answer.
4. Learn the six standard derivatives and write them from memory.

The modulus function

Cambridge insert: |x| as a graph, as an equation, as an inequality — and the reason it has no derivative at the corner.

LESSON 15–16 | 2 × 40 MIN | ALGEBRA 11, §1.5 (EXTENSION) | P1 · 1.6 · P2 · 1.1–1.3

LESSON CLOCK

8' | 14' | 17' | 18' | 13' | 10'

Three ways to read |x|	8 min
Graphs	14 min
Equations	17 min
Inequalities	18 min
The corner	13 min
Homework	10 min

LEARNING OBJECTIVES

- Define |x| piecewise and sketch y = |f(x)| and y = f(|x|).
- Solve equations containing a modulus, checking for false roots.
- Solve modulus inequalities by the two standard methods.
- Explain why |x| is not differentiable at zero.

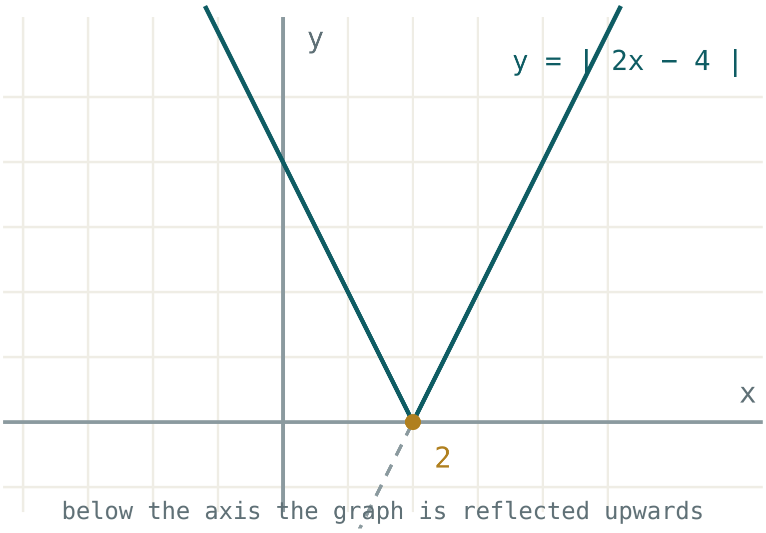
TEXTBOOK

National	pp. 75–80
Cambridge	Pure Mathematics 2 & 3, pp. 2–14
Workbook	P2 Exercise 1A–1C

01 Explanation

Three ways to read the same symbol

READING	STATEMENT
piecewise	$\lvert x\rvert = x \text{ if } x \geq 0,\ -x \text{ if } x < 0$
algebraic	$\lvert x\rvert = \sqrt{x^2}$
geometric	$\lvert x\rvert$ is the distance from x to 0; $\lvert a - b\rvert$ the distance between them



The V. Its two arms have gradient −1 and +1, and they meet at a corner.

THE GEOMETRIC READING SOLVES THE MOST

$|x-3| < 2$ says "x is less than 2 away from 3", so $1 < x < 5$ — no algebra needed at all.

Two different graphs

GRAPH	CONSTRUCTION	EFFECT
$y = \|f(x)\|$	reflect the part below the x-axis upwards	never negative
$y = f(\|x\|)$	keep $x \geq 0$, reflect it in the y-axis	always even

For $f(x) = x^2 - 4$: $|f(x)|$ has its dip between -2 and 2 flipped into a hump; $f(|x|)$ is unchanged, because f was already even.

THEY ARE DIFFERENT GRAPHS

Take $f(x) = x - 1$. Then $|f(x)|$ is a V with vertex $(1, 0)$, while $f(|x|)$ is a V with vertex $(0, -1)$ opening upwards. Read which one is asked for.

Equations

Two methods, both valid:

1. **Cases.** $|A| = B$ means $A = B$ or $A = -B$, provided $B \geq 0$.
2. **Squaring.** $|A| = |B| \iff A^2 = B^2$ — clean when both sides carry a modulus.

$$|2x-1| = 5 \Rightarrow 2x - 1 = 5 \text{ or } 2x - 1 = -5 \Rightarrow x = 3 \text{ or } x = -2$$

ALWAYS SUBSTITUTE BACK

$|x - 1| = 2x - 4$ gives $x = \frac{5}{3}$ and $x = 3$ by cases — but $\frac{5}{3}$ makes the right-hand side negative, and a modulus never is. Only $x = 3$ survives. Squaring is even more prone to this, because it destroys sign information.

Inequalities, and the corner

$$|A| < k \iff -k < A < k \quad |A| > k \iff A < -k \text{ or } A > k$$

The first is one double inequality; the second is genuinely two separate intervals and must be written with "or".

Why there is no derivative at the corner. For $f(x) = |x|$ at 0:

$$\frac{|0 + \mathrm{h}| - |0|}{\mathrm{h}} = \frac{|\mathrm{h}|}{\mathrm{h}} = +1\ (h > 0),\ -1\ (h < 0)$$

The two one-sided limits are $+1$ and -1. They disagree, so the limit does not exist and no tangent can be drawn. Everywhere else $|x|$ is perfectly differentiable — the failure is at exactly one point.

02 Worked examples

EXAMPLE 1 **Solve $|3x + 2| = 8$.**

1. $3x + 2 = 8 \Rightarrow x = 2$
2. $3x + 2 = -8 \Rightarrow x = -\frac{10}{3}$
3. Right-hand side is positive, so both are valid.

ANSWER

$x = 2$ or $x = -\frac{10}{3}$

EXAMPLE 2 **Solve $|x - 4| < 3$.**

1. $-3 < x - 4 < 3$
2. $1 < x < 7$
3. Geometrically: within 3 of 4.

ANSWER

$1 < x < 7$

EXAMPLE 3 **Solve $|2x - 1| = |x + 4|$.**

1. Square both sides.
   $(2x - 1)^2 = (x + 4)^2$
2. $4x^2 - 4x + 1 = x^2 + 8x + 16$
3. $3x^2 - 12x - 15 = 0 \Rightarrow x^2 - 4x - 5 = 0$
4. $x = 5$ or $x = -1$.
   Both check.

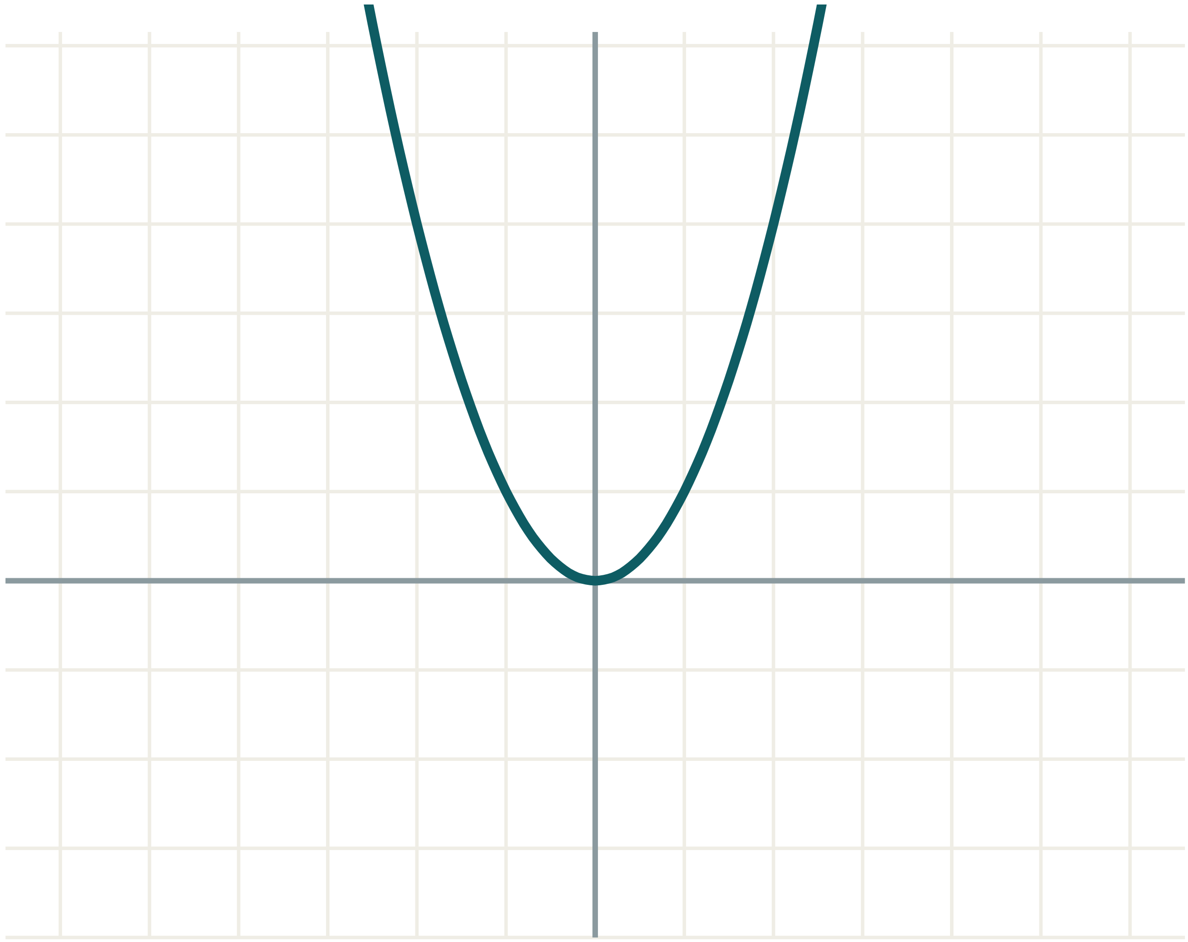
ANSWER

$x = 5$ or $x = -1$

03 Interactive model

Sketch the two sides as separate graphs and read the solutions off the intersections.

The V and its transformations



a **1** | b **0** | c **0** | equation `y = f(x)`

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Modulus (absolute value)	Modul	Модуль
Piecewise definition	Bo‘lakli ta’rif	Кусочное определение
Vertex of the modulus graph	Modul grafigi uchi	Вершина графика модуля
Critical value	Kritik qiymat	Критическое значение
False root	Yolg‘on ildiz	Посторонний корень
Squaring both sides	Ikkala tomonni kvadratga ko‘tarish	Возведение в квадрат
Distance on a number line	Sonlar o‘qidagi masofa	Расстояние на числовой прямой
Corner (cusp)	Sinish nuqtasi	Угловая точка
One-sided derivative	Bir tomonlama hosila	Односторонняя производная

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$|-7|$ is:

- -7
- 7
- 0
- undefined

$|x| < 4$ means:

- $x < 4$
- $-4 < x < 4$
- $x > -4$
- $x < -4$ or $x > 4$

$|x| > 4$ means:

- $-4 < x < 4$
- $x < -4$ or $x > 4$
- $x > 4$
- $x < 4$

$|x|$ at $x = 0$ has:

- derivative 0
- derivative 1
- no derivative
- derivative −1

$|x - 5| = 2$ gives:

- $x = 7$
- $x = 3$
- $x = 3$ or 7
- $x = \pm 2$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $|-9|$

ANSWER 9

2. $|0|$

ANSWER 0

3. Solve $|x| = 6$

ANSWER $x = \pm 6$

4. Solve $|x - 2| = 5$

ANSWER $x = 7, -3$

5. Solve $|x| < 3$

ANSWER $-3 < x < 3$

6. Solve $|x| \geq 2$

ANSWER $x \leq -2$ or $x \geq 2$

7. Vertex of $y = |x - 4|$

ANSWER $(4, 0)$

Medium · the standard question

7 PROBLEMS

1. Solve $|2x + 1| = 7$

ANSWER $x = 3, -4$

2. Solve $|3 - x| = 5$

ANSWER $x = -2, 8$

3. Solve $|x + 1| < 4$

ANSWER $-5 < x < 3$

4. Solve $|2x - 3| > 5$

ANSWER $x < -1$ or $x > 4$

5. Solve $|x - 1| = |x + 3|$

ANSWER $x = -1$

6. Sketch $y = |x^2 - 4|$ — where is the hump?

ANSWER between -2 and 2

7. Vertex of $y = |2x - 6|$

ANSWER $(3, 0)$

Hard · stretch and reason

7 PROBLEMS

1. Solve $|x - 1| = 2x - 4$

   ANSWER $x = 3$ only

2. Solve $|2x - 1| = |x + 4|$

   ANSWER $x = 5, -1$

3. Solve $|x^2 - 4| = 3$

   ANSWER $x = \pm 1, \pm\sqrt{7}$

4. Solve $|x - 2| + |x + 1| = 5$

   ANSWER $x = -2$ or $x = 3$

5. Solve $|x| + x = 6$

   ANSWER $x = 3$

6. Solve $|x - 3| < |x|$

   ANSWER $x > 1.5$

7. Show $|x|$ has one-sided derivatives ± 1 at 0

   ANSWER $\frac{|h|}{h}$ is $+1$ then -1

07 Homework

Homework — 5 tasks

Substitute every solution back. Modulus equations manufacture false roots.

1. Solve $|4x - 5| = 11$.
2. Solve $|x + 2| \le 6$ and $|3x - 1| > 8$.
3. Solve $|x - 3| = |2x + 1|$.
4. Sketch $y = |x^2 - 9|$ and $y = |x| - 3$ on separate axes.

5. Explain, with the two one-sided quotients written out, why $|x - 5|$ has no derivative at $x = 5$.

The equations of the tangent and the normal

Two lines at one point of a curve, from one derivative and one gradient relation.

LESSON 17–18 | 2 × 40 MIN | ALGEBRA 11, §1.6 | P1 · 7.6

LESSON CLOCK

8' | 16' | 16' | 17' | 13' | 10'

The three numbers you need	8 min
The tangent	16 min
The normal	16 min
Working backwards	17 min
Practice	13 min
Homework	10 min

LEARNING OBJECTIVES

- Write the equation of the tangent at a given point of a curve.
- Write the equation of the normal at that point.
- Find the point where a tangent has a given gradient.
- Find where a tangent meets the axes or the curve again.

TEXTBOOK

National	pp. 81–90
Cambridge	Pure Mathematics 1, pp. 159–167
Workbook	P1 Exercise 7F

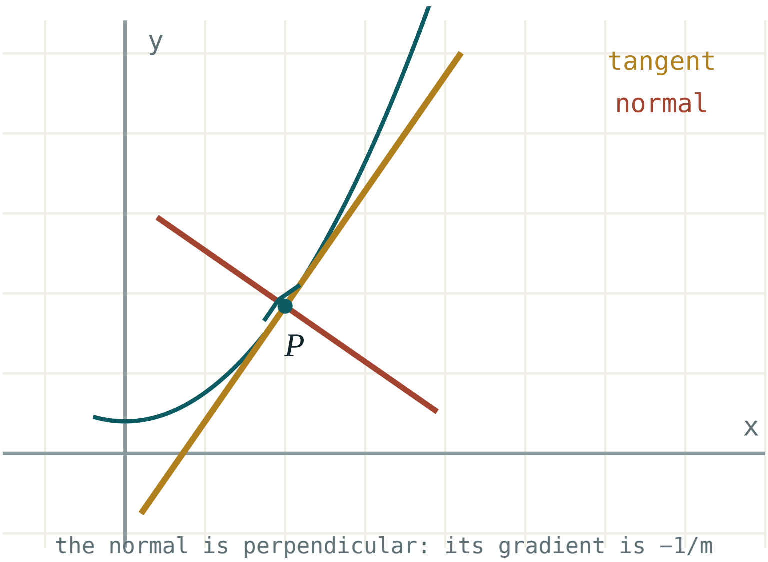
01 Explanation

Three numbers, then one formula

EVERY TANGENT QUESTION NEEDS THE SAME THREE

x_0 — where · $y_0 = f(x_0)$ — the point · $m = f'(x_0)$ — the gradient. Then the point-gradient form finishes it:

$$y - y_0 = m(x - x_0)$$



The tangent touches; the normal crosses at a right angle. Both pass through the same point.

DIFFERENTIATE BEFORE SUBSTITUTING

Substituting x_0 into f first leaves a constant, whose derivative is zero. Find $f'(x)$ in full, and only then put the number in.

The tangent

Example. Find the tangent to $y = x^2 - 3x$ at $x = 2$.

the point	$y = 4 - 6 = -2$, so $(2, -2)$
the gradient	$y' = 2x - 3$, so $m = 1$
the line	$y + 2 = 1(x - 2)$
simplified	$y = x - 4$

A **horizontal** tangent occurs where $f'(x) = 0$; a **vertical** one where f' fails to exist by becoming unbounded.

The normal

The **normal** is the line through the same point, perpendicular to the tangent:

$$m\_tangent \cdot m\_normal = -1 \Rightarrow m\_normal = -\frac{1}{f'(x_0)}$$

For the same curve at $(2, -2)$: the tangent gradient is 1, so the normal gradient is -1 and the normal is $y = -x$.

TWO SPECIAL CASES

If $f'(x_0) = 0$ the tangent is horizontal and the normal is **vertical**: $x = x_0$, which has no gradient at all. Writing $-\frac{1}{0}$ is not an answer.

Working backwards

Most examination questions give the gradient and ask for the point:

Example. Where does $y = x^3 - 4x$ have a tangent parallel to $y = 8x + 1$?

$$y' = 3x^2 - 4 = 8 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$$

Two points: $(2, 0)$ and $(-2, 0)$. Two tangents: $y = 8x - 16$ and $y = 8x + 16$.

PARALLEL AND PERPENDICULAR

Parallel to $y = mx + c$ means $f'(x) = m$. Perpendicular to it means $f'(x) = -\frac{1}{m}$. Both reduce to solving one equation.

02 Worked examples

EXAMPLE 1 **Find the tangent and the normal to $y = x^3 - 2x$ at $x = 1$.**

1. $y(1) = -1$

   Point $(1, -1)$.

2. $y' = 3x^2 - 2, y'(1) = 1$

3. Tangent: $y + 1 = 1(x - 1)$.

   $y = x - 2$

4. Normal: gradient -1.

   $y = -x$

ANSWER

tangent $y = x - 2$, normal $y = -x$

EXAMPLE 2 **Find the tangent to $y = \frac{4}{x}$ at $x = 2$, and where it meets the axes.**

1. $y(2) = 2$

   Point $(2, 2)$.

2. $y' = -\frac{4}{x^2}, y'(2) = -1$

3. Tangent $y = -x + 4$.

4. Axes: $(4, 0)$ and $(0, 4)$.

ANSWER

$y = 4 - x$; meets the axes at $(4, 0)$ and $(0, 4)$

EXAMPLE 3 **The tangent to $y = x^2$ at $x = a$ passes through $(0, -4)$. Find a.**

1. Point (a, a^2), gradient $2a$.
2. $y - a^2 = 2a(x - a)$
3. At $(0, -4)$: $-4 - a^2 = -2a^2$.
4. $a^2 = 4 \Rightarrow a = \pm 2$

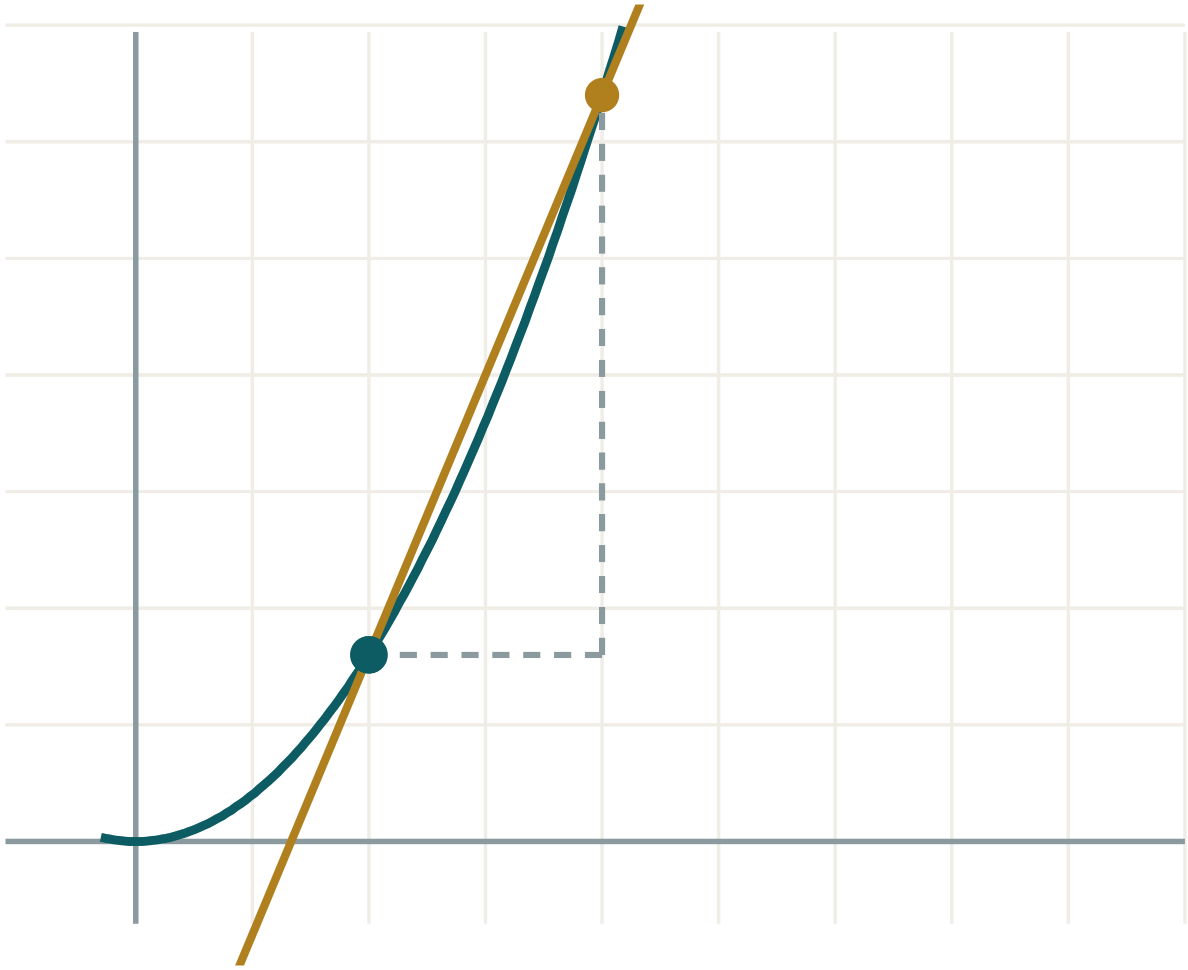
ANSWER

$a = \pm 2$

03 Interactive model

Draw both lines on one sketch — students who see only the tangent forget the normal is perpendicular.

Tangent at a point



x **2** | h **2** | Δy / Δx **2.4** | f ′(x) exactly **1.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Tangent	Urinma	Касательная
Normal	Normal	Нормаль
Point of tangency	Urinish nuqtasi	Точка касания
Gradient of a tangent	Urinma burchak koeffitsienti	Угловой коэффициент касательной
Perpendicular gradients	Perpendikulyar koeffitsientlar	Взаимно перпендикулярные
Point-gradient form	Nuqta-koeffitsient shakli	Уравнение через точку и угловой коэффициент
Horizontal tangent	Gorizontal urinma	Горизонтальная касательная
Intercept	Kesim	Отрезок на оси

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The gradient of the tangent at x_0 is:

- $f(x_0)$
- $f'(x_0)$
- $-\dfrac{1}{f'(x_0)}$
- 0

The normal gradient is:

- $f'(x_0)$
- $-\dfrac{1}{f'(x_0)}$
- $\dfrac{1}{f'(x_0)}$
- 0

A horizontal tangent occurs where:

- $f' = 1$
- $f' = 0$
- $f = 0$
- f' is undefined

If $f'(x_0) = 0$ the normal is:

- horizontal
- vertical
- $y = 0$
- undefined

Tangent to $y = x^2$ at $x = 3$:

- $y = 6x - 9$
- $y = 6x + 9$
- $y = 3x$
- $y = 9x - 6$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Tangent to $y = x^2$ at $x = 1$

   ANSWER $y = 2x - 1$

2. Normal to $y = x^2$ at $x = 1$

   ANSWER $y = -0.5x + 1.5$

3. Tangent to $y = 3x + 2$ at $x = 0$

   ANSWER $y = 3x + 2$

4. Gradient of the tangent to $y = x^3$ at $x = 2$

   ANSWER 12

5. Where is the tangent to $y = x^2 - 6x$ horizontal?

   ANSWER $x = 3$

6. Tangent to $y = x^2$ at $x = 0$

   ANSWER $y = 0$

7. Normal to $y = x^2$ at $x = 0$

   ANSWER $x = 0$

Medium · the standard question

7 PROBLEMS

1. Tangent to $y = x^3 - 2x$ at $x = 1$

   ANSWER $y = x - 2$

2. Normal to the same at the same point

   ANSWER $y = -x$

3. Tangent to $y = \frac{4}{x}$ at $x = 2$

   ANSWER $y = 4 - x$

4. Tangent to $y = \sqrt{x}$ at $x = 4$

   ANSWER $y = \frac{x}{4} + 1$

5. Where has $y = x^3 - 4x$ a tangent parallel to $y = 8x$

   ANSWER $x = \pm 2$

6. Where has $y = x^2 - 4x$ a tangent of gradient 2

   ANSWER $x = 3$

7. Tangent to $y = x^2 - 5x + 6$ at its x-intercepts

   ANSWER $y = -x + 2$ and $y = x - 3$

Hard · stretch and reason

7 PROBLEMS

1. Tangent to $y = x^2$ through $(0, -9)$

   ANSWER $y = \pm 6x - 9$

2. Normal to $y = x^2$ at $x = 1$ meets the curve again at

   ANSWER $x = -1.5$

3. Find k so $y = kx$ is tangent to $y = x^2 + 4$

   ANSWER $k = \pm 4$

4. Tangent to $y = \frac{1}{x}$ at $x = a$; show the axes cut a triangle of constant area

   ANSWER Area 2 always

5. Where do the tangents to $y = x^3$ at $x = 1$ and $x = -1$ meet?

   ANSWER $(0, -2)$

6. Normal to $y = \sqrt{x}$ at $x = 1$: find its equation

   ANSWER $y = -2x + 3$

7. Two curves $y = x^2$ and $y = \frac{1}{x}$ meet at $x = 1$. Find the angle between them

   ANSWER $\approx 71.6°$

07 Homework

Homework — 5 tasks

Give both lines in the form $y = mx + c$, simplified.

1. Find the tangent and the normal to $y = x^2 - 4x + 1$ at $x = 3$.
2. Find the tangent to $y = \frac{6}{x}$ at $x = 3$ and where it meets the axes.
3. Find the points on $y = x^3 - 3x^2$ where the tangent is horizontal.
4. Find where $y = x^3 + x$ has a tangent parallel to $y = 4x - 7$.

5. Find the tangent to $y = x^2$ that passes through $(0, -1)$.

Investigating a function with the derivative

Four lessons on the full procedure — sign of the derivative, stationary points, their nature, and the sketch that results.

LESSON 19–22 | 4 × 40 MIN | ALGEBRA 11, §1.7 | P1 · 7.7–7.8

LESSON CLOCK

14 | 21' | 27' | 27' | 35' | 36'

The sign of the derivative	14 min
Stationary points	21 min
Classifying them	27 min
The second derivative test	27 min
Full sketches	35 min
Practice and homework	36 min

LEARNING OBJECTIVES

- Use the sign of f′ to find intervals of increase and decrease.
- Find stationary points by solving f′(x) = 0.
- Classify each stationary point by the sign change or by f″.
- Sketch a curve using intercepts, stationary points and behaviour at the ends.

TEXTBOOK

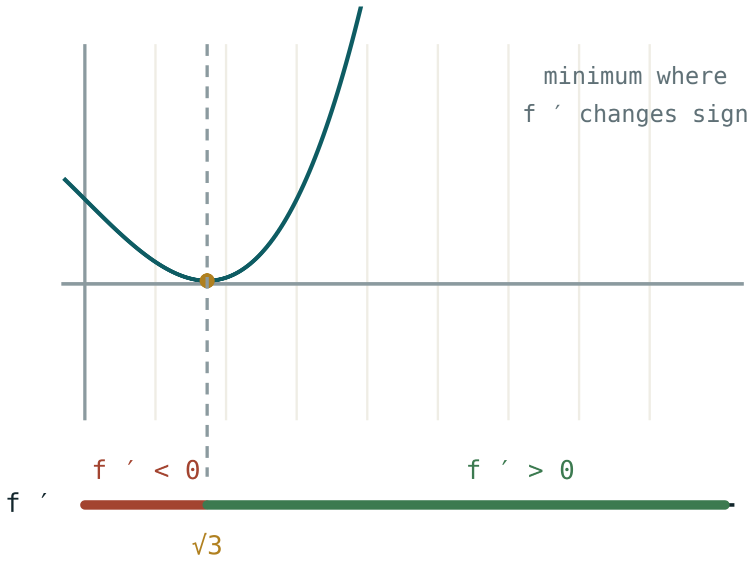
National	pp. 91–110
Cambridge	Pure Mathematics 1, pp. 168–185
Workbook	P1 Exercise 7G, 7H

01 Explanation

The sign of the derivative

THE LINK THAT MAKES CALCULUS USEFUL

$f'(x) > 0$ on an interval $\Longrightarrow$ f is **increasing** there. $f'(x) < 0 \Longrightarrow f$ is **decreasing**. $f'(x) = 0 \Longrightarrow$ the tangent is horizontal — a **stationary point**.



Where the derivative is above the axis, the curve climbs. Where it is below, the curve falls.

So the shape of f is read entirely off the **sign** of f', and the sign of f' is read off its factorised form.

Finding and tabulating

Example. $f(x) = x^3 - 3x^2 - 9x + 5$.

$$f'(x) = 3x^2 - 6x - 9 = 3(x - 3)(x + 1)$$

x	< -1	-1	-1 to 3	3	> 3
f'	$+$	0	$-$	0	$+$
f	↗	max	↘	min	↗

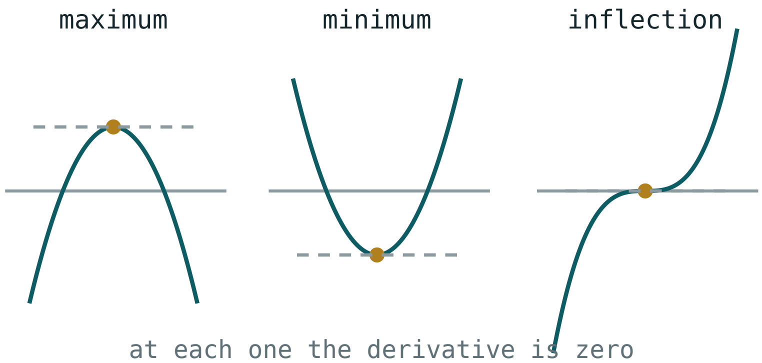
Local maximum at $(-1, 10)$, local minimum at $(3, -22)$.

ALWAYS FACTORISE f'

An unfactorised quadratic hides its sign. The factorised form gives the roots and the sign chart in one step.

Classifying: two tests

TEST	MAXIMUM	MINIMUM	INCONCLUSIVE WHEN
sign of f'	+ then −	− then +	never
$f''(x_0)$	$f'' < 0$	$f'' > 0$	$f'' = 0$



A maximum, a minimum and a stationary point of inflection. All three have a horizontal tangent.

$F''(X_0) = 0$ PROVES NOTHING

$y = x^4$ has $f''(0) = 0$ and a minimum; $y = x^3$ has $f''(0) = 0$ and an inflection. When the second-derivative test is silent, go back to the sign of f'.

A **stationary point of inflection** has $f' = 0$ but **no** sign change: the curve flattens and carries on in the same direction, as $y = x^3$ does at the origin.

The full sketch

SIX STEPS, IN ORDER

1 domain · **2** intercepts ($x = 0$, and $f(x) = 0$ if it factorises) · **3** f', factorised · **4** stationary points and their nature · **5** behaviour as $x \to \pm\infty$ · **6** draw, plotting the stationary points first.

For a cubic with a positive leading coefficient, step 5 says the curve comes up from $-\infty$ on the left and goes to $+\infty$ on the right. Together with the two stationary points, the shape is completely determined.

Second derivative and concavity. $f'' > 0$ means the curve bends upwards (concave up, holding water); $f'' < 0$ means it bends downwards. Where f'' changes sign there is a **point of inflection** — which need not be stationary.

02 Worked examples

EXAMPLE 1 **Investigate $f(x) = x^3 - 12x + 5$ fully.**

1. $f' = 3x^2 - 12 = 3(x - 2)(x + 2)$

2. Stationary at $x = \pm 2$.
   $f(-2) = 21$, $f(2) = -11$

3. $f'' = 6x$
   $f''(-2) = -12 < 0$ max; $f''(2) = 12 > 0$ min

4. Increasing on $x < -2$ and $x > 2$.
   Decreasing between.

ANSWER

max $(-2, 21)$, min $(2, -11)$

EXAMPLE 2 **Show $f(x) = x^3 + 3x$ has no stationary points.**

1. $f' = 3x^2 + 3$

2. $3x^2 + 3 = 0 \Rightarrow x^2 = -1$
   No real solution.

3. $f' > 0$ everywhere.

ANSWER

None — the function increases on all of $\mathbb{R}$

EXAMPLE 3 **Classify the stationary point of $f(x) = x^4 - 4x^3$ at $x = 0$.**

1. $f' = 4x^3 - 12x^2 = 4x^2(x - 3)$
   Stationary at 0 and 3.

2. $f'' = 12x^2 - 24x$, $f''(0) = 0$
   Test inconclusive.

3. Sign of f' near 0: $4x^2 \geq 0$, $(x - 3) < 0$.
   Negative on both sides.

4. No sign change.

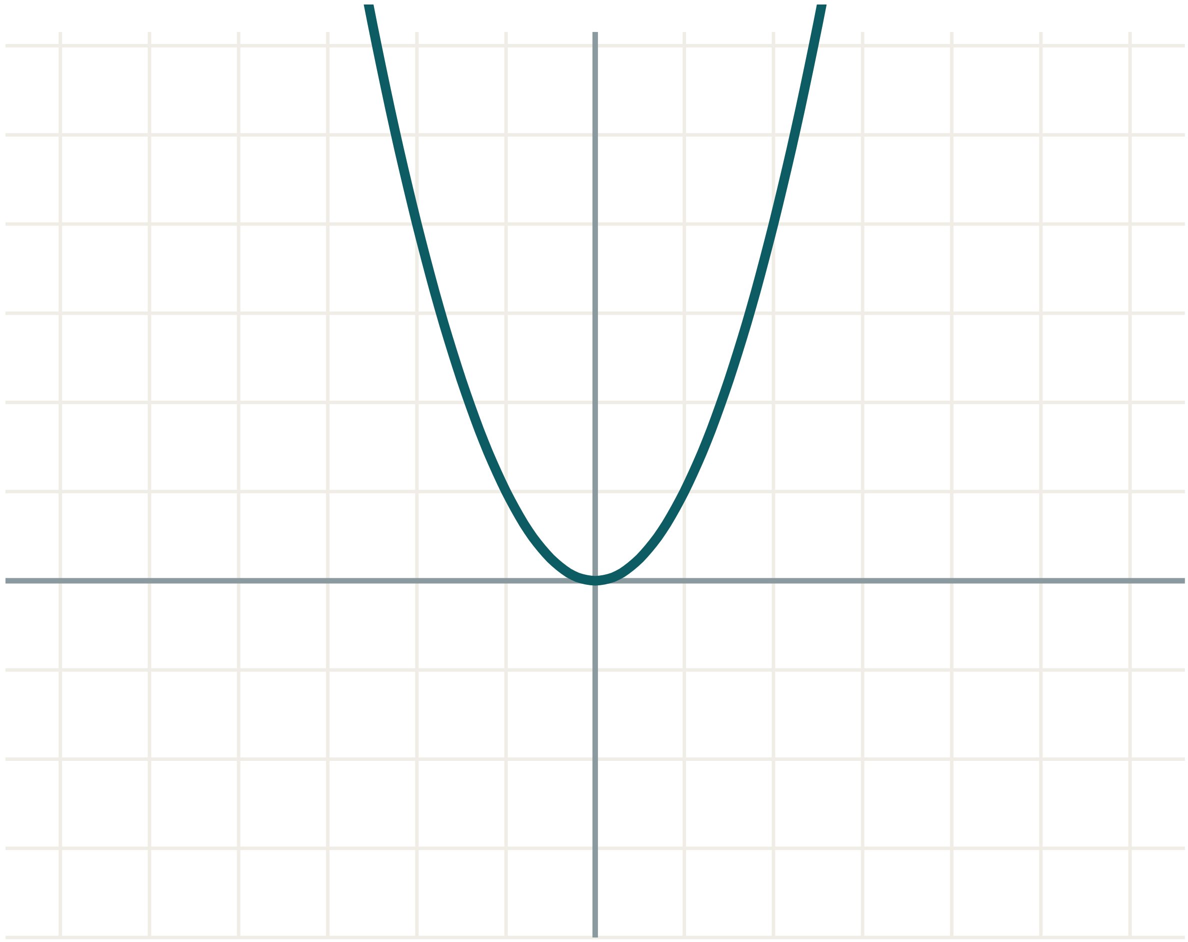
ANSWER

A stationary point of inflection at $(0, 0)$

03 Interactive model

Show the sign chart and the sketch side by side — every row of the table is a piece of the curve.

Shape from the derivative



a **1** | b **0** | c **0** | equation `y = f(x)`

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Increasing function	O‘suvchi funksiya	Возрастающая функция
Decreasing function	Kamayuvchi funksiya	Убывающая функция
Stationary point	Statsionar nuqta	Стационарная точка
Critical point	Kritik nuqta	Критическая точка
Local maximum	Lokal maksimum	Локальный максимум
Local minimum	Lokal minimum	Локальный минимум
Point of inflection	Egilish nuqtasi	Точка перегиба
Second derivative	Ikkinchi hosila	Вторая производная
Sign chart	Ishoralar jadvali	Таблица знаков
Curve sketching	Grafik chizish	Построение графика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$f'(x) > 0$ means the function is:

- positive
- increasing
- concave up
- stationary

A stationary point has:

- $f = 0$
- $f' = 0$
- $f'' = 0$
- $f' > 0$

$f' = 0$ and $f'' < 0$ gives a:

- minimum
- maximum
- inflection
- root

If $f''(x_0) = 0$ the point is:

- a maximum
- a minimum
- an inflection
- not decided by this test

Stationary points of $y = x^3 - 12x$ are at:

- $x = 0$
- $x = \pm 2$
- $x = \pm 4$
- $x = 12$

$y = x^3$ at $x = 0$ has:

- a maximum
- a minimum
- a stationary inflection
- no stationary point

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Stationary points of $y = x^2 - 6x$

   ANSWER $x = 3$

2. Classify it

   ANSWER minimum

3. Stationary points of $y = 4x - x^2$

   ANSWER $x = 2$, maximum

4. Is $y = 3x + 1$ increasing?

   ANSWER yes, everywhere

5. Stationary points of $y = x^3$

   ANSWER $x = 0$

6. Classify it

   ANSWER stationary inflection

7. f'' for $y = x^3 - 3x$

   ANSWER $6x$

Medium · the standard question

7 PROBLEMS

1. Investigate $y = x^3 - 3x$

   ANSWER max $(-1, 2)$, min $(1, -2)$

2. Investigate $y = x^3 - 12x + 5$

   ANSWER max $(-2, 21)$, min $(2, -11)$

3. Investigate $y = 2x^3 - 3x^2 - 12x$

   ANSWER max $(-1, 7)$, min $(2, -20)$

4. Where is $y = x^4 - 8x^2$ decreasing?

   ANSWER $x < -2$ and $0 < x < 2$

5. Stationary points of $y = x^4 - 8x^2$

   ANSWER $x = 0, \pm 2$

6. Nature of each

   ANSWER 0 max; ± 2 minima

7. Point of inflection of $y = x^3 - 6x^2$

   ANSWER $(2, -16)$

Hard · stretch and reason

7 PROBLEMS

1. Investigate $y = \dfrac{x}{x^2 + 1}$

   ANSWER max $(1, 0.5)$, min $(-1, -0.5)$

2. Investigate $y = x + \dfrac{1}{x}$

   ANSWER max $(-1, -2)$, min $(1, 2)$

3. Find a so $y = x^3 + ax$ has no stationary point

   ANSWER $a > 0$

4. Find a, b so $y = x^3 + ax^2 + b$ has a stationary point at $(2, 3)$

   ANSWER $a = -3, b = 7$

5. Classify the stationary point of $y = x^4$ at 0 and explain why f'' fails

   ANSWER Minimum; $f''(0) = 0$

6. Show $y = x^3 + x$ is increasing everywhere

   ANSWER $f' = 3x^2 + 1 > 0$

7. Sketch $y = x^3 - 3x^2 + 4$ fully

   ANSWER max $(0, 4)$, min $(2, 0)$, root at $x = -1$

07 Homework

Homework — 6 tasks

Every investigation needs the sign chart, not just the answers.

1. Investigate $y = x^3 - 6x^2 + 9x$ fully and sketch it.
2. Investigate $y = x^4 - 2x^2$ fully and sketch it.
3. Find the intervals on which $y = 2x^3 + 3x^2 - 36x$ increases.
4. Find and classify the stationary points of $y = x + \dfrac{4}{x}$.

5. Find a and b so that $y = x^3 + ax^2 + bx$ has stationary points at $x = 1$ and $x = 3$.

6. Explain, with two examples, why $f''(x_0) = 0$ does not classify a stationary point.

Extremum problems

The largest volume, the least cost, the shortest path — modelling first, then one derivative.

LESSON 23–25 | 3 × 40 MIN | ALGEBRA 11, §1.8 | P1 · 7.8

LESSON CLOCK

11' | 21' | 24' | 21' | 32' | 11'

The five-step method	11 min
Geometric problems	21 min
Boxes and cans	24 min
Cost and time problems	21 min
Practice	32 min
Homework	11 min

LEARNING OBJECTIVES

- Express the quantity to be optimised as a function of one variable.
- Use a constraint to eliminate the second variable.
- Find the extremum and justify that it is a maximum or a minimum.
- Interpret the answer, checking it against the physical domain.

TEXTBOOK

National	pp. 111–126
Cambridge	Pure Mathematics 1, pp. 178–190
Workbook	P1 Exercise 7H

01 Explanation

Five steps

THE METHOD, EVERY TIME

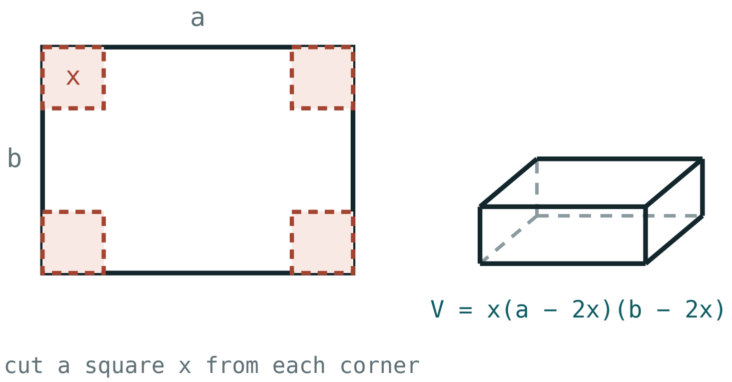
1 Name the variables and draw. **2** Write the quantity to optimise. **3** Use the constraint to reduce it to **one** variable, and state the feasible domain. **4** Differentiate, set to zero, solve. **5** Justify max or min, then answer in words with units.

STEP 3 IS WHERE MARKS ARE LOST

An expression in two variables cannot be differentiated. The constraint — the fixed perimeter, the fixed volume, the fixed length of wire — exists precisely to eliminate one of them.

The open box

Problem. A 24×24 cm square of card has equal squares of side x cut from its corners; the flaps fold up. Maximise the volume.



The card, the four corner squares removed, and the box that results.

$$V(x) = x(24 - 2x)^2 ,\ 0 < x < 12$$

$$V'(x) = (24 - 2x)^2 + x \cdot 2(24 - 2x)(-2) = (24 - 2x)(24 - 6x)$$

$V' = 0$ at $x = 12$ (rejected — outside the domain) and $x = 4$. Sign of V': positive before 4, negative after, so $x = 4$ is a maximum.

$$V(4) = 4 \times 16^2 = 1024\ cm^3$$

The cheapest can

Problem. A closed cylindrical can holds 500 cm³. Minimise the surface area.

$$V = \pi r^2 h = 500 \Rightarrow h = \frac{500}{\pi r^2}$$

$$S = 2\pi r^2 + 2\pi rh = 2\pi r^2 + \frac{1000}{r}$$

$$S' = 4\pi r - \frac{1000}{r^2} = 0 \Rightarrow r^3 = \frac{250}{\pi} \Rightarrow r \approx 4.30\ cm$$

Then $h \approx 8.60$ cm — exactly twice the radius. $S'' = 4\pi + \frac{2000}{r^3} > 0$, so it is a minimum. $S \approx 349$ cm².

THE RESULT IS WORTH REMEMBERING

For a closed cylinder of fixed volume, the cheapest shape always has $h = 2r$ — the height equals the diameter. Real cans are taller because the ends cost more per cm² than the wall.

Justifying, and the endpoints

A stationary point is not automatically the answer. Three ways to justify:

1. the sign of f' either side — always works;
2. f'' at the point — quick, but silent when it is zero;
3. compare with the **endpoint** values, when the domain is a closed interval.

ON A CLOSED INTERVAL, CHECK THE ENDS

The greatest value of a continuous function on $[a, b]$ occurs at a stationary point **or at an endpoint**. A problem asking for the maximum on $0 \le x \le 5$ needs $f(0)$ and $f(5)$ computed as well.

02 Worked examples

EXAMPLE 1 **A farmer has 100 m of fence for a rectangular pen against a straight wall. Maximise the area.**

1. Sides $x, x, 100 - 2x$.
   $0 < x < 50$
2. $A = x(100 - 2x) = 100x - 2x^2$
3. $A' = 100 - 4x = 0 \Rightarrow x = 25$
4. $A'' = -4 < 0$
   Maximum.

ANSWER

$1250\ m^2$, a 25 m by 50 m pen

EXAMPLE 2 **Find the point on $y = x^2$ closest to $(0, 3)$.**

1. $D^2 = x^2 + (x^2 - 3)^2$

   Minimise D^2, not D.

2. $= x^4 - 5x^2 + 9$

3. $(D^2)' = 4x^3 - 10x = 2x(2x^2 - 5) = 0$

4. $x = 0$ or $x = \pm\sqrt{2.5}$

   $x = 0$ gives $D = 3$; the others $D \approx 1.66$

ANSWER

$(\pm\sqrt{2.5}, 2.5)$, at distance $\dfrac{\sqrt{11}}{2} \approx 1.66$

EXAMPLE 3 **Find the greatest value of $f(x) = x^3 - 3x$ on $[0, 3]$.**

1. $f' = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$

   Only $x = 1$ is in range.

2. $f(1) = -2$

3. $f(0) = 0$, $f(3) = 18$

   Endpoints.

4. The largest is at an endpoint.

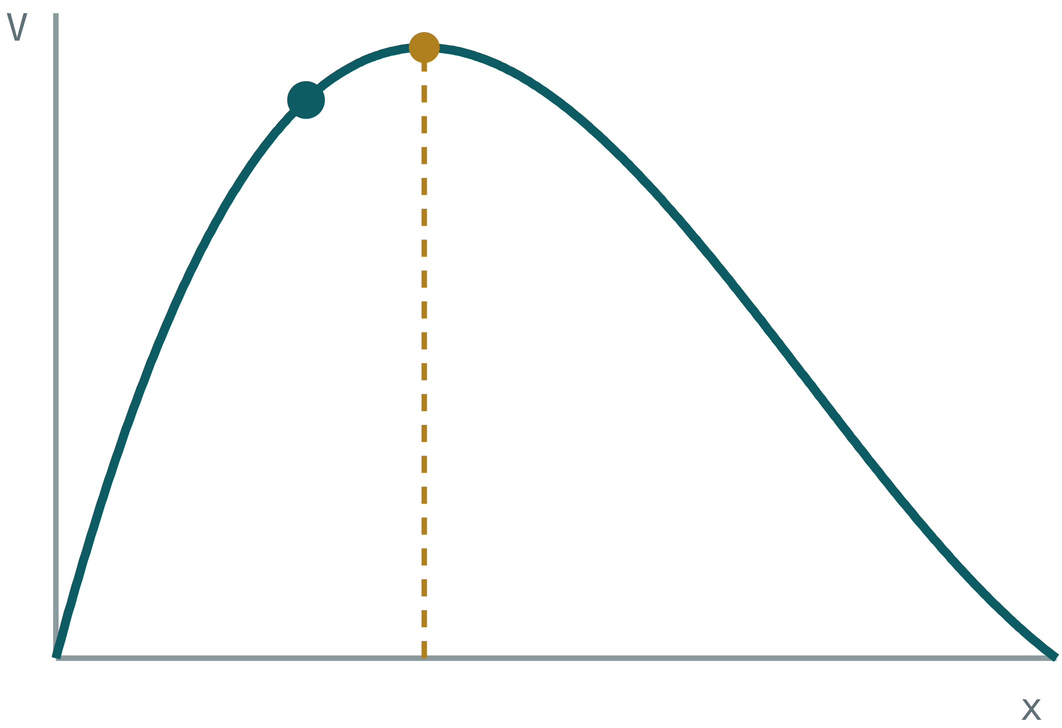
ANSWER

18, at $x = 3$

03 Interactive model

Slide x and watch the volume of the box rise and fall; mark the maximum.

The open box



x **2** | V(x) **384** | greatest V **420.1** | at x = **2.94**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Extremum	Ekstremum	Экстремум
Greatest value	Eng katta qiymat	Наибольшее значение
Least value	Eng kichik qiymat	Наименьшее значение
Constraint	Cheklov	Ограничение
Objective function	Maqsad funksiyasi	Целевая функция
Feasible domain	Ruxsat etilgan soha	Допустимая область
Endpoint value	Chegaraviy qiymat	Значение на конце
Optimisation	Optimallashtirish	Оптимизация
Justification	Asoslash	Обоснование

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The first thing to do in an extremum problem:

- differentiate
- express the quantity in one variable
- guess
- find f''

The constraint is used to:

- check the answer
- eliminate a variable
- find the domain only
- nothing

On a closed interval the greatest value can occur:

- only at a stationary point
- at a stationary point or an endpoint
- only at an endpoint
- anywhere

The cheapest closed cylinder of fixed volume has:

- $h = r$
- $h = 2r$
- $h = 4r$
- $r = 2h$

To minimise a distance it is easier to minimise:

- the distance
- its square
- its reciprocal
- its logarithm

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 100 m of fence, rectangular field. Maximum area

   ANSWER $625\ m^2$

2. Two numbers add to 20. Maximise the product

   ANSWER 100

3. Two numbers add to 16. Minimise the sum of squares

   ANSWER 128

4. Maximum of $y = 8x - x^2$

   ANSWER 16 at $x = 4$

5. Minimum of $y = x^2 - 10x$

   ANSWER -25 at $x = 5$

6. Domain for a corner cut x from a 20×20 card

   ANSWER $0 < x < 10$

7. Justify a maximum using f''

   ANSWER $f'' < 0$

Medium · the standard question

7 PROBLEMS

1. 80 m of fence against a wall. Maximum area

   ANSWER $800\ m^2$

2. 30×30 card, corner squares cut. Maximum volume

   ANSWER $2000\ cm^3$ at $x = 5$

3. Closed cylinder, volume 1000 cm³. Minimum surface area

   ANSWER $\approx 554\ cm^2$

4. Open cylinder (no lid), volume 500. Optimal radius

   ANSWER ≈ 5.42 cm

5. Greatest value of $y = x^3 - 3x^2$ on $[0, 4]$

   ANSWER 16 at $x = 4$

6. Least value of $y = x + \dfrac{9}{x}$ for $x > 0$

   ANSWER 6 at $x = 3$

7. Rectangle of perimeter 40 with maximum area

   ANSWER 100, a 10-square

Hard · stretch and reason

7 PROBLEMS

1. A rectangle inscribed in a circle of radius 5. Maximise the area

   ANSWER 50, a square of side $5\sqrt{2}$

2. Point on $y = x^2$ closest to $(0, 3)$

   ANSWER $(\pm\sqrt{2.5}, 2.5)$

3. A cone of slant height 6. Maximise its volume

   ANSWER $r = 2\sqrt{6}$, $V = 16\pi\sqrt{3}$

4. Wire of 60 cm cut into a square and a circle. Minimise the total area

   ANSWER Square ≈ 33.6 cm of wire

5. Cheapest open-topped box of square base and volume 32 cm³

   ANSWER base 4 cm, height 2 cm

6. A window is a rectangle topped by a semicircle, perimeter 10 m. Maximise the area

   ANSWER $r = \dfrac{10}{\pi + 4} \approx 1.40\ m$

7. A man rows at 4 km/h and walks at 5 km/h from a point 3 km offshore to a point 8 km along. Minimise the time

   ANSWER Land 4 km along the shore

07 Homework

Homework — 6 tasks

Every answer needs the justification and the units.

1. 120 m of fence, rectangular pen against a wall. Find the greatest area.
2. A 40×40 cm card: corner squares are cut and the flaps folded. Find the greatest volume.
3. A closed cylindrical can holds 750 cm³. Find the radius that minimises the surface area, and check $h = 2r$.

4. Find the greatest and least values of $y = x^3 - 6x^2 + 9x$ on $[0, 4]$.
5. Find the point on $y = \sqrt{x}$ closest to $(4, 0)$.
6. Two numbers have a sum of 30. Maximise the product of one and the square of the other.

Control work 2, and the quarter review

The whole derivative block in one paper, then the map that connects a sign chart to a sketch to an optimum.

LESSON 26–27 | 2 × 40 MIN | ALGEBRA 11, NAZORAT ISHI 2 | P1 · CHAPTER 7 REVIEW

LESSON CLOCK

3 | 36' | 9' | 18' | 11' | 3

Instructions	3 min
The paper	36 min
Answers on the board	9 min
Rewrite	18 min
Concept map	11 min
Targets	3 min

LEARNING OBJECTIVES

- Apply the whole quarter in one assessment.
- Choose the correct rule and the correct test without prompting.
- Build a concept map of the derivative and its applications.
- Set a personal target for Quarter II.

TEXTBOOK

National	pp. 127–130
Cambridge	Pure Mathematics 1, pp. 191–194
Workbook	Control paper B

01 Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Differentiate $y = x^4 - \dfrac{3}{x^2} + 2\sqrt{x}$	3	L7–9
2	Differentiate $y = x^2(3x - 1)^4$ and factorise	4	L10–12
3	Solve $\lvert 2x - 5\rvert \le 7$ and $\lvert x - 1\rvert = \lvert 3x + 2\rvert$	4	L15–16
4	Tangent and normal to $y = x^3 - 2x^2$ at $x = 2$	5	L17–18
5	Investigate $y = x^3 - 3x^2 - 9x + 2$ and sketch it	7	L19–22
6	A 36×36 card, corner squares cut: maximise the volume, with justification	7	L23–25

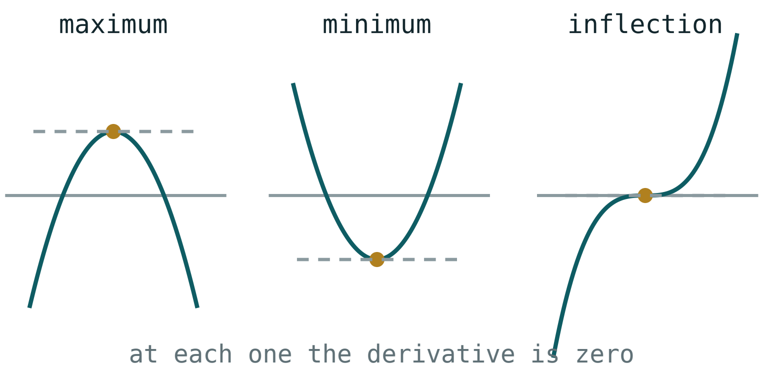
Q5 AND Q6 CARRY THE SIGN CHART

Q5 gives 3 marks for the stationary points and 4 for the chart, the nature and the sketch. Q6 gives 2 of its 7 for the justification alone. An unjustified optimum is an incomplete answer.

The concept map

Six boxes, links written as sentences:

- **increment → limit** — "shrink Δx and the quotient settles"
- **limit → derivative** — "the derivative *is* that limit"
- **derivative → rules** — "power, sum, product, quotient, chain"
- **derivative → tangent and normal** — "gradient at a point"
- **sign of f' → shape** — "positive climbs, negative falls, zero is stationary"
- **shape → optimisation** — "the extremum is where the gradient turns"



Three stationary points — the picture that links the sign chart to the sketch.

Looking forward

Quarter II opens with approximation by the tangent and with the indefinite integral — differentiation run backwards. Every rule of this quarter is used in reverse, so a rule half learned now costs twice as much then.

ONE HABIT TO CARRY FORWARD

Factorise f' before doing anything with it. Every sign chart, every stationary point, every optimisation in Quarter II starts from the factorised derivative.

02 Worked examples

EXAMPLE 1 **Model answer, Q5: investigate $y = x^3 - 3x^2 - 9x + 2$.**

1. $y' = 3x^2 - 6x - 9 = 3(x - 3)(x + 1)$

2. Stationary at $x = -1, 3$.

   $y(-1) = 7$, $y(3) = -25$

3. $y'' = 6x - 6$

   $y''(-1) = -12$ max; $y''(3) = 12$ min

4. Increasing outside $[-1, 3]$, decreasing inside.

ANSWER

max $(-1, 7)$, min $(3, -25)$

EXAMPLE 2 **Model answer, Q6: 36×36 card, corner squares of side x.**

1. $V = x(36 - 2x)^2$

   $0 < x < 18$

2. $V' = (36 - 2x)(36 - 6x)$

3. $x = 6$ (or 18, rejected).

4. $V' > 0$ before 6, $V' < 0$ after.

   Maximum.

ANSWER

$V = 6 \times 24^2 = 3456\ cm^3$

EXAMPLE 3 **Model answer, Q2: differentiate $y = x^2(3x-1)^4$.**

1. Product, with the chain rule inside.
2. $y' = 2x(3x-1)^4 + x^2 \cdot 4(3x-1)^3 \cdot 3$
3. $= 2x(3x-1)^3[(3x-1) + 6x]$
4. $= 2x(3x-1)^3(9x-1)$

ANSWER

$2x(3x-1)^3(9x-1)$

03 Interactive model

Work Q5 on the board with the sign chart drawn under the sketch.

The quarter in ten questions

$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$:

- 0
- 2
- 4
- undefined

First principles, $(x^3)'$:

- $3x$
- $3x^2$
- x^2
- $3x^3$

$\left(\frac{1}{x^2}\right)'$:

- $-\frac{2}{x^3}$
- $\frac{2}{x^3}$
- $-\frac{1}{x^3}$
- $-2x$

$((3x-1)^4)'$:

- $4(3x-1)^3$
- $12(3x-1)^3$
- $3(3x-1)^3$
- $12(3x-1)^4$

Solve $|2x-5| \le 7$:

- $x \le 6$
- $-1 \le x \le 6$
- $x \ge -1$
- $x \le -1$ or $x \ge 6$

Tangent to $y = x^3 - 2x^2$ at $x = 2$ has gradient:

- 0
- 4
- 8
- 12

Stationary points of $y = x^3 - 3x^2 - 9x$:

- $x = 0, 3$
- $x = -1, 3$
- $x = 1, -3$
- $x = \pm 3$

$f' = 0$, $f'' > 0$ means:

- maximum
- minimum
- inflection
- nothing

36×36 card: optimal corner cut is:

- 4
- 6
- 9
- 12

On a closed interval the maximum can be at:

- a stationary point only
- an endpoint only
- either
- neither

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Concept map	Tushunchalar xaritasi	Карта понятий
Command word	Topshiriq so‘zi	Командное слово
Justification	Asoslash	Обоснование
Self-assessment	O‘z-o‘zini baholash	Самооценка
Target	Maqsad	Цель
Revision	Takrorlash	Повторение
Summative assessment	Yakuniy baholash	Итоговое оценивание

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

“Show that” requires:

- the answer
- every line
- a sketch
- a calculator

An optimisation answer without justification loses:

- nothing
- a mark or two
- all marks
- half

Before using a sign chart you should:

- sketch
- factorise f'
- find f''
- substitute

Quarter II begins with:

- trigonometry
- the integral
- probability
- vectors

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $(x^4)'$

ANSWER $4x^3$

2. $\left(\frac{3}{x^2}\right)'$

ANSWER $-\frac{6}{x^3}$

3. $(2\sqrt{x})'$

ANSWER $\frac{1}{\sqrt{x}}$

4. Solve $|2x - 5| \le 7$

ANSWER $-1 \le x \le 6$

5. Gradient of $y = x^3 - 2x^2$ at $x = 2$

ANSWER 4

6. Stationary points of $y = x^2 - 4x$

ANSWER $x = 2$

7. 36×36 card: best corner cut

ANSWER 6

Medium · the standard question

7 PROBLEMS

1. $\left(x^4 - \dfrac{3}{x^2} + 2\sqrt{x}\right)'$

ANSWER $4x^3 + \dfrac{6}{x^3} + \dfrac{1}{\sqrt{x}}$

2. $(x^2(3x - 1)^4)'$

ANSWER $2x(3x - 1)^3(9x - 1)$

3. Solve $|x - 1| = |3x + 2|$

ANSWER $x = -0.25$ or $x = -1.5$

4. Tangent to $y = x^3 - 2x^2$ at $x = 2$

ANSWER $y = 4x - 8$

5. Normal there

ANSWER $y = -0.25x + 0.5$

6. Investigate $y = x^3 - 3x^2 - 9x + 2$

ANSWER max $(-1, 7)$, min $(3, -25)$

7. 36×36 card: greatest volume

ANSWER $3456\ cm^3$

Hard · stretch and reason

7 PROBLEMS

1. $\left(\dfrac{x^2}{\sqrt{x + 1}}\right)'$

ANSWER $\dfrac{x(3x + 4)}{2(x + 1)\text{^}(3/2)}$

2. Find a, b so $y = x^3 + ax^2 + bx$ has a max at $x = -1$ and min at $x = 2$

ANSWER $a = -1.5, b = -6$

3. Solve $|x - 2| + |x + 3| = 7$

ANSWER $x = -4$ or $x = 3$

4. Tangent to $y = x^3$ at $x = a$ meets the curve again where?

ANSWER $x = -2a$

5. Closed cylinder volume 2000 cm^3: minimise surface area

ANSWER $r \approx 6.83$ cm

6. Greatest value of $y = x^2(6 - x)$ on $[0, 6]$

ANSWER 32 at $x = 4$

7. Show that between two roots of a differentiable function there is a stationary point

ANSWER Rolle's theorem — the curve must turn

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter II.

1. Rewrite in full every control-work question that lost a mark.
2. Finish the concept map with all six links written as sentences.
3. Investigate $y = x^3 - 6x^2 + 9x + 1$ fully and sketch it.
4. Write your target for Quarter II in one checkable sentence, and date it.